

Polylogarithms: Uniform Bounds and Structural Transitions

Sharp Euler constants, Möbius depth, radial degeneracies,
and a square-root scale for reflected log-gamma coefficients

A research continuation prepared for Vladimir Reshetnikov
ProveIt research programme

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Abstract

We continue the current ProveIt manuscript along four concrete open directions. First, a sharp two-point inequality for the Euler remainder kernel proves one-sided enclosures for every real outer order $a \geq 0$ and inner order $b > 0$. The optimal uniform constant is the previously identified Gaussian axis maximum $C_* = \max_{b>0} \{\beta(b) + 2^{-b}\eta(b)\}$; the new result is the remainder comparison that makes this magnitude extremum an error constant. An exact finite certificate gives $C_* < 57/50$. A positive gamma-series identity yields a moving-order phase law and the asymptotically optimal global scaled-error constant one, with an explicit lower asymptotic for its approach. Second, a binary free-Lie decomposition and confluent interpolation compute the formal depth reached by any prescribed number of projective letter directions, including the six Möbius transformations. Certificates for an established Nielsen reduction and a corresponding weight-eight obstruction illustrate the distinction between a formal certificate and a numerical period relation. Third, rational interval arithmetic isolates a simultaneous quadratic and quartic radial degeneracy and proves a negative sixth coefficient. The resulting local families exhibit a quarter-power birth of a maximum and a region with a minimum followed by a maximum. Finally, a uniform saddle analysis proves Gaussian damping when the reflected exponent is of order the square root of the late coefficient index, together with a growing-exponent half-term truncation law. A shifted saddle extends the coefficient result to a small proportional range and resolves a further cubic transition. Proofs, exact certificates, diagnostic computations, proposed manuscript changes, and a focused research agenda are supplied. No numerical residual is promoted to an identity, and no claim of numerical period independence or global priority is made.

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Primary manuscript. Analysis/Polylogarithms/docs/manuscript/polylogarithms.tex.

Proof status. Ordinary mathematical proofs and exact finite arithmetic certificates; no proof-assistant formalization.

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1 Scope, baseline, and the new results

The linked repository contains a consolidated manuscript, not merely the earlier collection of experimental identities. At the pinned revision it has already incorporated proofs of the Gaussian weight-six double identities, the displayed weight-four triples, the mixed S_4 relation, several rank formulas, and the cubic and quartic trilogarithm ladders. It also contains sharp real-order kernel theorems, integral distribution modules, reflected moments, and a substantial zero-geometry programme. Those advances must be treated as the starting point. Repeating them would not answer the request for a continuation.

We audited the active questions against the source chapters and the later reports preserved beside them. Four directions admit substantive progress with proofs that can be stated precisely and checked independently. They share a method: first find the exact object that controls the question, and then separate its analytic or algebraic structure from numerical diagnostics. For Euler errors that object is a two-point kernel difference. For depth it is the osculating span of binary forms. For radial transitions it is a convergent implicit equation, together with certified low-order derivatives. For reflected moments it is a saddle integral with an oscillatory amplitude and a signed cut representation.

1.1 A theorem and status ledger

Direction	Status in the audited material	Contribution of this article
Euler errors	Order-dependent bounds; sharp critical constant; axis magnitude maximum; unrestricted remainder comparison open.	A sharp all-order remainder theorem, its optimal global constant, an exact rational budget, positive moment identities, and a moving-order phase law with an asymptotically optimal constant and an explicit lower asymptotic for its approach.
Formal depth	Exact calculations, including a weight-six obstruction and special transformations.	An all-weight formula for any number of projective directions; six-map versus full-linear comparison; exact certificates for known Nielsen reductions, with literature attribution.
Radial geometry	Both signs of the quartic coefficient on the quadratic threshold; a diagnostic simultaneous zero.	Exact isolation, negative sixth coefficient, a quarter-power law, a two-extremum local unfolding, and a fully explicit rational example.
Reflected coefficients	Fixed reflected exponent asymptotics and fixed-exponent optimal truncation.	A uniform $m = O(\sqrt{k})$ transition, Gaussian damping, explicit first correction, a growing-exponent half-term law, and a small proportional saddle theorem.
Special-value candidates	S_6 and the revised S_8 candidate remain conjectural despite certified small residuals.	Their evaluation status is unchanged; the new formal obstruction specifies a limitation of a particular reduction method.

The labels “new” and “open” in this article refer to the audited repository state. The general tools—Euler transformation, gamma integrals, free-Lie character theory, Hermite interpolation, analytic implicit inversion, and the saddle method—are classical. We do not claim that a broad literature search could establish priority for every consequence. The article supplies proofs of the stated refinements and records the primary sources used for their background.

1.2 Conventions and the interpretation of identities

Multiple polylogarithms use decreasing nested indices:

$$\mathrm{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z_1^{n_1} \dots z_d^{n_d}}{n_1^{s_1} \dots n_d^{s_d}}$$

where the series converges. A one-variable specialization means $\mathrm{Li}_{s_1, \dots, s_d}(z, 1, \dots, 1)$. Real-order $F_{a,b}$ is defined explicitly in Section 2, so no convention for nonintegral-depth words is needed. All analytic continuations use the principal branch unless a simply connected path domain is specified. Statements at i concern analytic or Abel values where the ordinary Fourier series fails to converge.

Formal identities in a shuffle quotient have a different meaning. Their vanishing proves an exact congruence modulo the stated product and endpoint-constant relations. A nonzero class proves the absence of a certificate using those relations. It does not prove numerical independence of periods and does not exclude other functional equations. Section 3 gives the exact comparison space before drawing any depth conclusion.

For parameter estimates, terminating decimal endpoints are rational numbers. They can therefore be part of a finite exact proof when every operation is outward rounded and every transcendental input has a proved remainder bound. By contrast, a printed approximate root or a many-digit agreement from quadrature is diagnostic unless an enclosure theorem is also supplied.

1.3 What is reproducible

The archive includes this source and its compiled PDF, independent exact verifiers for the finite claims, numerical checks that compare different representations, and the resulting machine-readable data. The new statements are proved in the text; the code has three different roles, explicitly separated in the verification section: finite arithmetic proof certificates, finite corroboration of a general algebraic theorem, and nonrigorous asymptotic diagnostics.

The repository was read at the pinned revision without applying changes to it. Suggested chapter insertions and wording replacements are supplied separately. The package can thus be reviewed and integrated against a concrete baseline without silently altering historical reports or their recorded status.

2 A sharp uniform Euler bound at every positive real order

The canonical manuscript proves one-sided Euler enclosures in several parameter regions. It also proves that the Gaussian magnitude has a unique global maximum on the outer-order axis. It explicitly cautions that a magnitude bound alone is not a remainder bound in the supercritical region. The missing implication is supplied here by a sharp, elementary two-point inequality. No finite signed-moment representation is needed.

2.1 Definitions and the exact two-measure representation

For $a \geq 0$, $b > 0$, let

$$H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad F_{a,b}(z) = \sum_{n \geq 1} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad |z| < 1.$$

For integral positive orders this is the decreasing-index multiple polylogarithm $\text{Li}_{a,b}(z, 1)$. Write $g_{a,b} = \Im F_{a,b}(i)$ for the principal analytic value, and set

$$f_n = \frac{H_{2n}^{(b)}}{(2n+1)^a}, \quad d_j = \sum_{n=0}^j (-1)^n \binom{j}{n} f_n, \quad E_N = \sum_{j=0}^{N-1} \frac{d_j}{2^{j+1}} \quad (N \geq 1). \quad (2.1)$$

Thus $f_0 = d_0 = E_1 = 0$. The sign convention for d_j is $(-1)^j$ times the usual forward difference. Euler's classical transformation is recalled in DLMF [2]; our task is to prove its continuation, sign, and sharp uniform error throughout the present real-order family.

For $s > 0$, define a probability measure on $(0, 1)$ by

$$d\sigma_s(t) = \frac{(-\log t)^{s-1}}{\Gamma(s)} dt, \quad \sigma_0 = \delta_1.$$

Its moments are $\int_0^1 t^{n-1} d\sigma_s(t) = n^{-s}$. Writing the inner harmonic number as a geometric sum gives

$$\frac{H_{n-1}^{(b)}}{n^a} = \iint t^{n-1} \frac{1 - u^{n-1}}{1 - u} d\sigma_a(t) d\sigma_b(u). \quad (2.2)$$

This includes $n = 1$, when the integrand is zero. For every fixed n it is bounded, so no cancellation of divergent integrals is involved.

Lemma 2.1 (Two-measure continuation). *For $a \geq 0$, $b > 0$, the identity*

$$F_{a,b}(z) = z^2 \iint \frac{t}{(1 - tz)(1 - tuz)} d\sigma_a(t) d\sigma_b(u) \quad (2.3)$$

holds on $\mathbb{C} \setminus [1, \infty)$. In particular,

$$-g_{a,b} = \iint \frac{t^2(1+u)}{(1+t^2)(1+t^2u^2)} d\sigma_a(t) d\sigma_b(u) > 0. \quad (2.4)$$

For $j \geq 1$,

$$d_j = - \iint \frac{(1 - t^2u^2)^j - (1 - t^2)^j}{1 - u} d\sigma_a(t) d\sigma_b(u) < 0. \quad (2.5)$$

Proof. Insert (2.2) in the disk series and sum the two geometric series. Absolute convergence on compact subdisks justifies the interchange. On a compact subset of the slit plane, both denominators in (2.3) are bounded away from zero uniformly for $t, u \in [0, 1]$. The integral is therefore holomorphic there and gives the asserted continuation. Rationalizing at i proves (2.4). Applying the finite binomial theorem to (2.2) proves (2.5). The numerator is strictly positive for $t > 0$, $0 < u < 1$, giving the sign. $\square$

2.2 The sharp two-point kernel inequality

Put

$$W_N(x) = \frac{(1-x)^N}{1+x}, \quad 0 \leq x \leq 1.$$

The two arguments of W_N will be $t^2 u^2$ and t^2 .

Lemma 2.2 (Power-difference maximum). *Let $0 < v < 1$ and let $k \geq 2$ be an integer. Then*

$$\begin{aligned} \max_{0 \leq x \leq 1} \{(1-vx)^k - (1-x)^k\} &= \frac{(1-v)^k}{(1-v^{k/(k-1)})^{k-1}} \\ &\leq \frac{1-v}{1+v}. \end{aligned} \quad (2.6)$$

The second inequality is an equality when $k = 2$, and is strict when $k > 2$.

Proof. Differentiation shows that the unique interior maximizer satisfies $(1-x)/(1-vx) = v^{1/(k-1)}$. Substitution gives the displayed maximum. For $1 \leq p \leq 2$, the function

$$H(p) = \log(1-v^p) - \log(1-v) - (p-1)\log(1+v)$$

has $H(1) = H(2) = 0$ and

$$H''(p) = -\frac{v^p(\log v)^2}{(1-v^p)^2} < 0.$$

Thus $1-v^p \geq (1-v)(1+v)^{p-1}$, with strict inequality for $1 < p < 2$. Take $p = k/(k-1)$ and raise to the power $k-1$. $\square$

Theorem 2.3 (Sharp Euler kernel modulus). *For $N \geq 1$, $0 \leq x \leq 1$, and $0 < v < 1$,*

$$0 \leq W_N(vx) - W_N(x) \leq \frac{1-v}{1+v}. \quad (2.7)$$

The upper bound is attained exactly when $N = 1$ and $x = 1$. For $N \geq 2$ it is strict for every x .

Proof. The nonnegativity follows from the decrease of W_N . For $N = 1$,

$$W_1(vx) - W_1(x) = \frac{2x(1-v)}{(1+x)(1+vx)}.$$

The comparison with $(1-v)/(1+v)$ is equivalent to $(1-x)(1-vx) \geq 0$, with equality only at $x = 1$. For $N \geq 2$, the uniformly convergent geometric expansion

$$W_N(x) = \sum_{k=N}^{\infty} 2^{N-k-1} (1-x)^k \quad (2.8)$$

expresses W_N as a probability mixture of powers. Apply Lemma 2.2 term by term. At least one index $k > 2$ has positive mixture weight, so the upper inequality is strict. $\square$

This proof also explains why a direct estimate by a signed measure's total variation can be wasteful: the relevant object is a difference of two values of the remainder kernel, and this difference has a uniform modulus even where a precursor measure has infinite mass.

2.3 Uniform continuation and the least possible constant

Let $\beta(s) = \sum_{n \geq 0} (-1)^n / (2n+1)^s$ and $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} / n^s$ for $s > 0$, and define

$$C(b) = \beta(b) + 2^{-b} \eta(b) = \int_0^1 \frac{1+u}{1+u^2} d\sigma_b(u). \quad (2.9)$$

The equality follows either from the residue classes in $\text{Li}_b(i)$ or from (2.4) at $a = 0$. The measure integral also avoids any question about conditional termwise summation.

Theorem 2.4 (All-order Euler enclosure). *For every $a \geq 0$, $b > 0$, the finite sums E_N decrease strictly to $g_{a,b}$, with*

$$R_N(a, b) := 2^N (E_N - g_{a,b}) = \iint \frac{W_N(t^2 u^2) - W_N(t^2)}{1 - u} d\sigma_a(t) d\sigma_b(u). \quad (2.10)$$

For all $N \geq 1$,

$$0 < R_N(a, b) \leq C(b) < \frac{1 + \sqrt{2}}{2} < \frac{5}{4}. \quad (2.11)$$

Equality $R_N(a, b) = C(b)$ holds exactly when $a = 0$ and $N = 1$. Consequently, for each fixed $b > 0$, $C(b)$ is the least error constant uniform in $a > 0$ and $N \geq 1$. Furthermore, $R_N(a, b) \rightarrow 0$ for every fixed pair.

Proof. For fixed t, u , summing the finite geometric series in (2.5) and comparing with (2.4) gives (2.10). All finite integrals exist: a difference quotient at $u = 1$ has a finite continuous limit. Alternatively, the bound below gives direct domination of the remainder. Apply Theorem 2.3 with $v = u^2$, $x = t^2$ and divide by $1 - u$:

$$0 \leq \frac{W_N(t^2 u^2) - W_N(t^2)}{1 - u} \leq \frac{1 + u}{1 + u^2}.$$

Integration proves the bound and the equality conditions, since σ_b has positive density on $(0, 1)$ and σ_a does too when $a > 0$. The maximum of $(1 + u)/(1 + u^2)$ occurs at $u = \sqrt{2} - 1$ and equals $(1 + \sqrt{2})/2$; positivity of the density makes the corresponding integral inequality strict. The strict decrease of E_N follows from $d_N < 0$. The bound proves convergence to the analytic value even if the original boundary series diverges. Finally the integrand tends to zero at every $t, u > 0$ with $u < 1$, so dominated convergence gives $R_N \rightarrow 0$.

For sharpness at fixed b , let $a \downarrow 0$. If $T_a \sim \text{Gamma}(a, 1)$, then $\mathbb{E}T_a = a$, so $e^{-T_a} \rightarrow 1$ in probability. Thus $\sigma_a \Rightarrow \delta_1$. At $N = 1$, dominated convergence gives $R_1(a, b) \rightarrow C(b)$. No smaller constant can hold uniformly over positive a . $\square$

Corollary 2.5 (Sharp global constant). *Let b_* denote the unique maximizer of C on $(0, \infty)$, and set*

$$C_* = C(b_*) = \max_{b > 0} \{\beta(b) + 2^{-b} \eta(b)\}. \quad (2.12)$$

Then the sharp uniform strict enclosure for $a, b > 0$ is

$$\boxed{E_N - C_* 2^{-N} < g_{a,b} < E_N \quad (N \geq 1).} \quad (2.13)$$

It is sharp also over the entire nonnegative outer-order domain, where the non-strict lower enclosure is attained only at $(a, b, N) = (0, b_, 1)$. The exact bounds from the manuscript are*

$$1 < b_* < 2, \quad \frac{227}{200} < C_* < \frac{1 + \sqrt{2}}{2}.$$

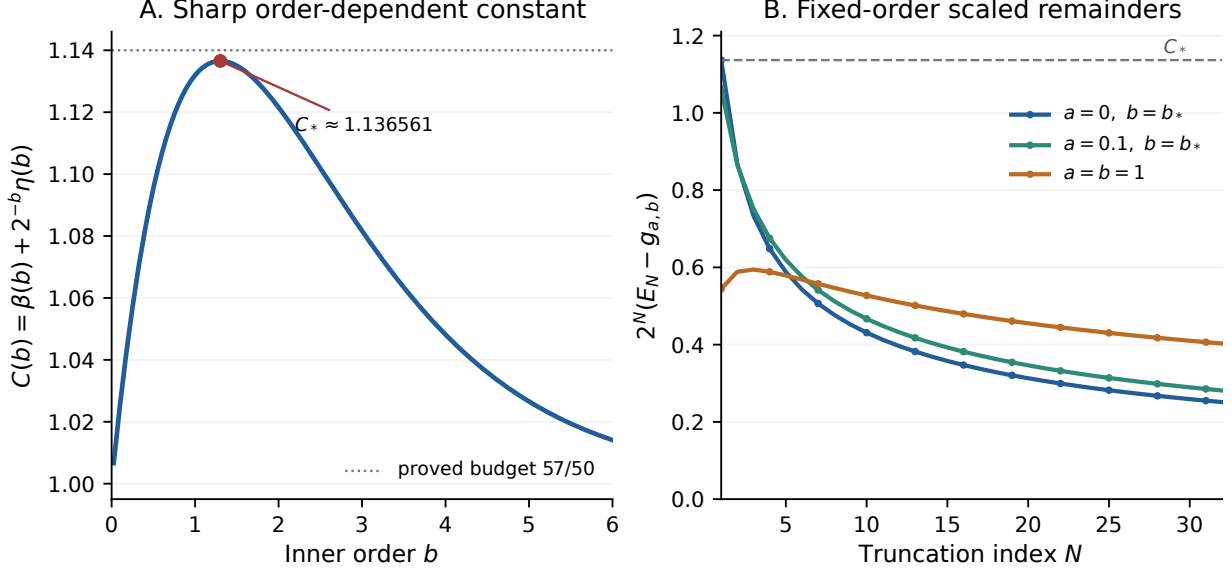


Figure 1: Numerical illustrations of the proved Euler bounds. Panel A shows $C(b)$, its maximum, and the rational budget $57/50$. Panel B shows fixed-order scaled remainders, including the increase from $N = 1$ to $N = 2$ at $a = b = 1$. The curve at $a = 0$ is the boundary case allowed by the theorem. The plots illustrate the analytic proof; the rational global budget is certified separately.

The diagnostic numerical values are

$$b_* = 1.30221658710124120959\dots, \quad C_* = 1.13656110333950956095\dots$$

Proof. The axis maximum, its uniqueness and its nondegeneracy are proved in the pinned manuscript, 05-subcritical-stieltjes.tex, Theorem subpick:thm:axis-maximum [1]. For clarity, the mechanism is that $C(b) = \mathbb{E}[\phi(T_b)]$, where $\phi(T) = (1 + e^{-T})/(1 + e^{-2T})$ has exactly one maximum. The strict two-intersection property of the Mellin family proves uniqueness; this is stronger than a numerical root scan. The new inequality (2.11) supplies the missing remainder comparison. Maximizing it over b gives the claimed bound. Its sharpness follows from $N = 1$, $b = b_*$, and $a \downarrow 0$. $\square$

The constant is specified exactly by (2.12); the decimal display is not an asserted closed form or a certified enclosure. The elementary rational constant $5/4$ already follows from the analytic proof. The following small exact computation improves the operational constant uniformly over all orders.

Proposition 2.6 (A simple rational global budget). *The exact inequalities*

$$C_* < \frac{5129}{4500} < \frac{57}{50}$$

hold. Thus $57/50$ is a valid uniform Euler error budget for every $a \geq 0$, $b > 0$, and $N \geq 1$.

Proof. For $1 \leq b \leq 2$, gamma differentiation and Cauchy–Schwarz give

$$|C'(b)| = |\text{Cov}(\log T_b, \phi(T_b))| \leq \sqrt{\psi_1(b)} \sqrt{\text{Var } \phi(T_b)} < \frac{1}{6}.$$

Indeed $1 \leq \phi \leq (1 + \sqrt{2})/2$ has range less than $1/4$, so its standard deviation is less than $1/8$. The variance identity $\text{Var} \log T_b = \psi_1(b)$ follows by differentiating the gamma integral. Also $\psi_1(b) \leq \psi_1(1) = \pi^2/6 < 16/9$; the last inequality follows, for example, from $\pi < 22/7$.

The standard-library verifier encloses $C(1 + j/30)$ for all $0 \leq j \leq 30$ by (2.16) with $a = 0$, $N = 48$, using a 10^{45} grid and integer root inequalities. The already proved $5/4$ tail bound makes these genuine analytic enclosures. All thirty-one upper endpoints are strictly less than $1137/1000$. Every point of $[1, 2]$ is within $1/60$ of a node, whence

$$C(b) < \frac{1137}{1000} + \frac{1}{360} = \frac{5129}{4500} < \frac{57}{50}.$$

The proved location $1 < b_* < 2$ completes the global argument. Every finite comparison is recorded as a rational inequality in `data/euler_exact_certificate.json`. Neither floating-point root finding nor a numerical derivative bound enters this proof. $\square$

Remark 2.7 (Why the new kernel step is necessary). At $(a, b) = (1, 1)$, the identity $F_{1,1}(z) = \frac{1}{2} \log^2(1 - z)$ gives $g_{1,1} = -\pi \log 2/8$. Therefore

$$R_1 = \frac{\pi \log 2}{4}, \quad R_2 = \frac{\pi \log 2}{2} - \frac{1}{2} > R_1.$$

Indeed $\pi > 3$ and $\log 2 > 2/3$ imply $\pi \log 2 > 2$. Thus the scaled errors need not decrease outside the previously proved right-endpoint region. Bounding R_N by its $N = 1$ value would be invalid even at this simplest integral-order pair.

2.4 A general theorem for harmonic sums built from moments

The argument proves a more general statement that can be reused for other completely monotone inner sequences. Let α, β be finite positive measures on $[0, 1]$, and define

$$c_n = \iint t^{n-1} (1 + u + \cdots + u^{n-2}) d\alpha(t) d\beta(u), \quad c_1 = 0.$$

Use the definitions (2.1) with $f_n = c_{2n+1}$, and let g be the imaginary value at i of the continuation (2.3). Difference quotients at $u = 1$ are understood by continuity.

Theorem 2.8 (Moment-product Euler principle). *For every such pair of measures and every $N \geq 1$,*

$$0 \leq 2^N (E_N - g) \leq \alpha([0, 1]) \int_0^1 \frac{1+u}{1+u^2} d\beta(u) \leq \frac{1+\sqrt{2}}{2} \alpha([0, 1]) \beta([0, 1]). \quad (2.14)$$

The last universal constant is optimal. For probability measures it is attained at $N = 1$, $\alpha = \delta_1$, $\beta = \delta_{\sqrt{2}-1}$.

Proof. The finite identities and kernel comparison above remain valid for arbitrary positive measures. Passing to $u = 1$ preserves the inequality by continuity. The endpoint $t = 0$ contributes zero, so it causes no difficulty. Integrate the pointwise bound, then optimize its last factor on $[0, 1]$. The stated two Dirac masses attain it. $\square$

This theorem identifies the property required for an extension: the coefficients must be a product of a Hausdorff moment and a finite partial sum of another moment sequence. It does not assert the same bound for an arbitrary alternating series or for signed input measures.

2.5 A certified algorithm and exact identity family

For every $a \geq 0$, $b > 0$, the following absolutely convergent identity is now valid even when the untransformed Gaussian boundary series fails the term test:

$$g_{a,b} = \sum_{j=1}^{\infty} \frac{1}{2^{j+1}} \sum_{n=1}^j (-1)^n \binom{j}{n} \frac{H_{2n}^{(b)}}{(2n+1)^a}. \quad (2.15)$$

The terms of the outer series are strictly negative, and the first $N-1$ terms have error bounded by $C(b)2^{-N}$. This is a proved identity family with an explicit error theorem, rather than a new name for a formal divergent rearrangement.

The already available finite binomial-tail formula is

$$E_N = 2^{-N} \sum_{n=0}^{N-1} (-1)^n \frac{H_{2n}^{(b)}}{(2n+1)^a} \sum_{j=n+1}^N \binom{N}{j}. \quad (2.16)$$

The new theorem and Proposition 2.6 supply the order-independent tail budget $(57/50)2^{-N}$. For rational a, b , fractional powers in this finite sum can be enclosed by integer root inequalities; additions and multiplications then use rational interval arithmetic. This avoids the parameter-dependent singular budgets near $a+b=1$ in the earlier implementation. Given a target absolute radius ε , any N satisfying $(57/50)2^{-N} \leq \varepsilon$ suffices for the analytic tail, in addition to the separately controlled finite-sum rounding.

The accompanying program checks this statement using exact rational arithmetic, including deliberately corrupted candidate controls. Its numerical comparisons use different representations for selected known values; a finite check corroborates the analytic proof but does not replace it. The new global theorem should supplement the more specialized and sometimes sharper bounds already present in the book.

2.6 A positive gamma-series identity for the scaled remainder

The same representation gives an exact identity that exposes the effect of a large inner order. Let T, V be independent gamma variables of rate one and respective shapes a, b ; for $a=0$ take $T=0$ deterministically. Define

$$J_N(s) = \mathbb{E}[W_N(e^{-2Y_s})], \quad Y_s \sim \text{Gamma}(s, 1), \quad J_N(0) = W_N(1) = 0.$$

In particular, for $s > 0$ this is the positive integral

$$J_N(s) = \frac{1}{\Gamma(s)} \int_0^\infty y^{s-1} e^{-y} \frac{(1 - e^{-2y})^N}{1 + e^{-2y}} dy.$$

Proposition 2.9 (Positive gamma-series identity). *For all $a \geq 0$, $b > 0$ and integers $N \geq 1$,*

$$R_N(a, b) = \sum_{r=1}^{\infty} r^{-b} \mathbb{E} \left[W_N(e^{-2(T+V/r)}) - W_N(e^{-2T}) \right]. \quad (2.17)$$

Every summand is nonnegative, and the series converges. If $b > 1$, its first term gives the uniform enclosure

$$0 \leq R_N(a, b) - \{J_N(a+b) - J_N(a)\} \leq \zeta(b) - 1. \quad (2.18)$$

Proof. In logarithmic coordinates the exact two-measure formula reads

$$R_N(a, b) = \mathbb{E} \frac{W_N(e^{-2(T+U)}) - W_N(e^{-2T})}{1 - e^{-U}}, \quad U \sim \text{Gamma}(b, 1),$$

with T, U independent. Expand $(1 - e^{-U})^{-1}$ as a positive geometric series. Tonelli's theorem applies because the numerator is nonnegative. Multiplying the rate-one gamma density of U by $e^{-(r-1)U}$ produces r^{-b} times the density of V/r . This proves (2.17); its convergence follows from Theorem 2.4. The sum of independent rate-one gamma variables has the sum of their shapes, so the $r = 1$ term is $J_N(a + b) - J_N(a)$. Every other expectation lies in $[0, 1]$, which proves (2.18). $\square$

The enclosure is independent of a and N . For large b , it replaces a two-variable integral by a difference of two positive one-variable integrals with an exponentially small budget in b . For $b \leq 1$ the positive series identity still holds, but the crude tail estimate by $\zeta(b) - 1$ must not be used.

2.7 Logarithmically growing orders: a phase law for Euler errors

The fixed-order statement $R_N \rightarrow 0$ cannot be made uniform over all positive orders. The following theorem identifies an explicit transition and its two boundary profiles. Write

$$\Phi_G(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy$$

for the standard normal distribution function.

Theorem 2.10 (Moving-order Euler phase law). *Let $\ell = \log N$. Suppose*

$$a_N = A\ell + \alpha\sqrt{\ell} + o(\sqrt{\ell}), \quad b_N = B\ell + \beta\sqrt{\ell} + o(\sqrt{\ell}), \quad A, B > 0,$$

where α, β are fixed real numbers. Then

$$R_N(a_N, b_N) \rightarrow \begin{cases} 0, & A + B < 1/2, \\ \Phi_G(\sqrt{2}(\alpha + \beta)), & A + B = 1/2, \\ 1, & A < 1/2 < A + B, \\ \Phi_G(-\sqrt{2}\alpha), & A = 1/2, \\ 0, & A > 1/2. \end{cases} \quad (2.19)$$

The cases are disjoint because $B > 0$, and cover the stated parameter range. In particular,

$$\liminf_{N \rightarrow \infty} \sup_{a, b > 0} R_N(a, b) \geq 1.$$

Proof. Since $b_N \rightarrow \infty$, (2.18) shows that

$$R_N(a_N, b_N) = J_N(a_N + b_N) - J_N(a_N) + o(1).$$

For example $\zeta(b) - 1 \leq 2^{-b} + 2^{1-b}/(b-1)$ for $b > 1$, by an integral comparison, so the discarded term tends to zero. It remains to evaluate $J_N(s_N)$ for

$$s_N = c\ell + \gamma\sqrt{\ell} + o(\sqrt{\ell}), \quad c > 0.$$

The function $W_N(e^{-2y})$ is an approximate threshold at $y = \ell/2$. Set $h_N = \ell^{1/4}$. If $y \leq \ell/2 - h_N$, then

$$0 \leq W_N(e^{-2y}) \leq \exp(-e^{2h_N}).$$

If $y \geq \ell/2 + h_N$, Bernoulli's inequality gives

$$0 \leq 1 - W_N(e^{-2y}) \leq (N+1)e^{-2y} \leq 2e^{-2h_N}.$$

Thus the expectation of its difference from $\mathbf{1}_{\{Y_{s_N} > \ell/2\}}$ tends to zero provided

$$\mathbb{P}(|Y_{s_N} - \ell/2| \leq h_N) \rightarrow 0.$$

When $c \neq 1/2$, this follows from $Y_{s_N}/\ell \rightarrow c$ in probability, since $\text{Var } Y_{s_N} = s_N = O(\ell)$. When $c = 1/2$, it follows from the gamma central limit theorem and $h_N/\sqrt{s_N} \rightarrow 0$. For completeness, that central limit theorem follows directly from the characteristic function

$$\mathbb{E}e^{it(Y_s - s)/\sqrt{s}} = e^{-it\sqrt{s}}(1 - it/\sqrt{s})^{-s} \longrightarrow e^{-t^2/2}.$$

The limiting normal distribution is continuous, so intervals whose standardized widths tend to zero have probability tending to zero, including at the moving centers here.

Consequently

$$J_N(s_N) \longrightarrow \begin{cases} 0, & c < 1/2, \\ \Phi_G(\sqrt{2}\gamma), & c = 1/2, \\ 1, & c > 1/2. \end{cases}$$

Apply this first to $s_N = a_N + b_N$ and then to $s_N = a_N$. Their difference gives all five cases in (2.19), using $1 - \Phi_G(x) = \Phi_G(-x)$. Choosing, for example, $A = B = 1/3$ proves the final assertion. $\square$

The phase law supplies the lower bound in an asymptotically sharp global result.

Theorem 2.11 (Asymptotically optimal global constant). *Let*

$$C_N = \sup_{a, b > 0} 2^N (E_N(a, b) - g_{a, b}).$$

Then

$$\lim_{N \rightarrow \infty} C_N = 1. \quad (2.20)$$

The same conclusion holds if $a = 0$ is included in the supremum.

Proof. Put $F_N(y) = W_N(e^{-2y})$. This function is increasing from zero to one and satisfies $0 \leq F'_N(y) \leq 4$ uniformly in $N \geq 1$, $y \geq 0$. Indeed, with $x = e^{-2y}$,

$$F'_N(y) = \frac{2x(1-x)^{N-1}\{N+1+(N-1)x\}}{(1+x)^2}.$$

For $N \geq 2$ it is at most $4Nx(1-x)^{N-1} \leq 4$; for $N = 1$ it is at most one. For every finite L , $\sup_{0 \leq y \leq L} F'_N(y) \rightarrow 0$.

Let $M(a)$ be the maximum of the density of Gamma($a, 1$) for $a > 1$. Conditional on $U = u$, positivity of F'_N and $\int_0^\infty F'_N = 1$ give

$$\mathbb{E}_T[F_N(T+u) - F_N(T)] = \int_0^u \mathbb{E}_T F'_N(T+s) ds \leq uM(a).$$

Since $u/(1 - e^{-u}) \leq 1 + u$ for $u \geq 0$, it follows that

$$R_N(a, b) \leq M(a)(1 + b). \quad (2.21)$$

Moreover $M(a) \rightarrow 0$ as $a \rightarrow \infty$. An elementary quantitative bound suffices: for $L = a - 1 \geq 4$, the gamma density has its maximum at L , and on $|y - L| \leq \sqrt{L}/2$,

$$\log \frac{p_a(y)}{p_a(L)} = L \log \left(1 + \frac{y - L}{L} \right) - (y - L) \geq -\frac{1}{4},$$

using $\log(1 + z) \geq z - z^2$ for $|z| \leq 1/2$. Integrating on that interval yields $M(a) \leq e^{1/4}/\sqrt{a-1}$.

We also need uniform convergence on bounded order rectangles:

$$\sup_{0 \leq a \leq A_0, 0 < b \leq B_0} R_N(a, b) \longrightarrow 0 \quad (A_0, B_0 < \infty). \quad (2.22)$$

For any fixed pair in the rectangle, couple independent gamma variables so that $T \leq T_0$ and $U \leq U_0$, where T_0, U_0 have shapes A_0, B_0 and rate one. The gamma addition rule supplies this coupling; shape zero is a point mass at zero. The exact kernel is bounded by

$$\frac{F_N(T + U) - F_N(T)}{1 - e^{-U}} \leq (1 + U_0) \sup_{0 \leq y \leq T_0 + U_0} F'_N(y).$$

The last random variable is independent of the smaller orders, is bounded by $4(1 + U_0)$, and tends to zero almost surely. Dominated convergence proves (2.22).

Now fix $\varepsilon > 0$. Choose $B_0 > 1$ so large that $\zeta(B_0) < 1 + \varepsilon$. For $b > B_0$, (2.18) gives $R_N(a, b) \leq \zeta(b) < 1 + \varepsilon$, uniformly in a, N . For $b \leq B_0$, choose A_0 sufficiently large that (2.21) is less than ε for all $a > A_0$. The remaining compact rectangle is controlled by (2.22). Thus $\limsup C_N \leq 1 + \varepsilon$. Let $\varepsilon \downarrow 0$. The choice $a_N = b_N = (\log N)/3$ in Theorem 2.10 gives $\liminf C_N \geq 1$. This proves (2.20), also with the axis included. $\square$

The global constant $C_* > 1$ and the asymptotic constant one answer different questions. The first is sharp uniformly over all truncation indices; the second is the limit of the optimal order-uniform constant at index N . The proof of the latter does not give an effective rate for $C_N - 1$, nor does it assert that C_N decreases in N .

2.8 An explicit lower asymptotic for the optimal constant

For the scaled Euler remainder $R_N(a, b) = 2^N(E_N - g_{a,b})$, write $C_N = \sup_{a,b>0} R_N(a, b)$. Values at $a = 0$ give lower bounds for this supremum by continuity as $a \downarrow 0$. The positive integral representation gives

$$R_N(0, b) = \int_0^1 K_N(u) d\sigma_b(u), \quad K_N(u) = \frac{(1+u)(1-u^2)^{N-1}}{1+u^2}.$$

Proposition 2.12 (Explicit axis asymptotics). *Put*

$$p = \frac{\log 2}{\log(3/2)}, \quad q = \frac{\log 3}{\log 2}, \quad c = (1 - q^{-1})q^{-p}, \quad \widehat{b}_N = \frac{\log(Nq)}{\log(3/2)}.$$

For every integer $N \geq 1$ and $b > 0$,

$$\begin{aligned} R_N(0, b) &= 1 + 2^{-b} - N3^{-b} - N4^{-b} + \varepsilon_N(b), \\ 0 \leq \varepsilon_N(b) &\leq \frac{N^2 - N + 2}{2} (5^{-b} + 6^{-b}). \end{aligned}$$

Consequently,

$$R_N(0, \widehat{b}_N) = 1 + cN^{-p} + o(N^{-p}), \quad \liminf_{N \rightarrow \infty} N^p(C_N - 1) \geq c.$$

Proof. Let $F_N(s) = (1-s)^{N-1}/(1+s)$. Then $F_N(0) = 1$, $F'_N(0) = -N$, and $0 \leq F''_N(s) \leq N^2 - N + 2$ on $[0, 1]$. For $N \geq 3$ the last assertion follows term by term from

$$F''_N(s) = \frac{(N-1)(N-2)(1-s)^{N-3}}{1+s} + \frac{2(N-1)(1-s)^{N-2}}{(1+s)^2} + \frac{2(1-s)^{N-1}}{(1+s)^3}.$$

For $N = 1, 2$ it follows directly from $F_1(s) = (1+s)^{-1}$ and $F_2(s) = 2(1+s)^{-1} - 1$. Taylor's integral remainder therefore gives

$$0 \leq F_N(s) - 1 + Ns \leq \frac{N^2 - N + 2}{2} s^2.$$

Multiply by $1+u$, set $s = u^2$, and use $\int u^j d\sigma_b(u) = (j+1)^{-b}$. This proves the explicit remainder bounds.

The function $2^{-b} - N3^{-b}$ has its unique maximum at $b = \hat{b}_N$, where its value is cN^{-p} . At that value of b , the omitted terms are bounded in magnitude by

$$O\left(N^{1-2p} + N^{2-\log 5/\log(3/2)}\right) = o(N^{-p}).$$

Indeed $p > 1$, and $\log 5/\log(3/2) > p + 2$ is equivalent to $5 > 9/2$. This proves the asymptotic identity and the lower bound for C_N . $\square$

Numerically, $p = 1.70951129135145477698\dots$ and $c = 0.16794779654251279566\dots$. The lower bound is a theorem; the matching upper asymptotic over all positive orders is the separate conjecture in Section 7.

3 Exact depth bounds under the six standard argument changes

The existing continuation proves that, at weight six, products and the six transformations permuting $0, 1, \infty$ leave exactly one formal triple direction outside the span of doubles. It also observes that the same weight-six span is obtained from all invertible linear changes of the two differential forms. We extend both calculations to every weight and depth. The equality of these two spans at weight six does not persist: at weight seven their dimensions are respectively 12 and 14.

The height-one application belongs to the established theory of Nielsen polylogarithms. Charlton–Gangl–Radchenko give its polynomial model and depth bound in [3, §3.3, Theorem 7]; their Proposition 28 is exactly the weight-seven congruence below. We supply a formal membership proof, an explicit weight-eight Lie certificate, and a fully replayable product completion of that known congruence. The principal extension here is the rank formula for the full binary shuffle quotient, including every irreducible component and any finite set of directions.

All lower bounds in this section concern the specified formal shuffle presentation. They do not assert numerical independence at $z = i$, and they impose no obstruction on a proof using additional cyclotomic letters, distribution, stuffle, or other functional equations.

3.1 The precise presentation

Let $V = \mathbb{Q}x \oplus \mathbb{Q}y$ and let $\text{Sh}(V)$ be the shuffle algebra on binary words. Write J for its augmentation ideal, and set

$$Q_w = (J/J^2)_w, \quad D_{w,d} = \text{span}\{[v] : |v| = w, v \text{ contains at most } d \text{ copies of } y\}.$$

Here $[v]$ denotes the class modulo shuffle products of nonempty words. We use the full binary shuffle algebra so that every linear letter substitution acts immediately. At weights $w \geq 2$, adding words

ending in x does not enlarge the indecomposables: shuffle normalization with $H_x(z) = \log z$ removes trailing x letters.

The analytic interpretation is outer letter first:

$$x = \frac{dt}{t}, \quad y = \frac{dt}{1-t}, \quad H_{x^{s_1-1}y \dots x^{s_r-1}y}(z) = \text{Li}_{s_1, \dots, s_r}(z, 1, \dots, 1).$$

Pure x words have the standard logarithmic regularization. Products are assigned the sum of the weights of their factors.

The two pullback matrices

$$R = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \quad (3.1)$$

act on columns. Thus $R(x) = -y$, $R(y) = -x$, while $M(x) = x + y$, $M(y) = -y$. They correspond to $z \mapsto 1 - z$ and $z \mapsto z/(z - 1)$. The group

$$\mathfrak{S} = \{I, R, M, RM, MR, RMR\}$$

has six elements. It sends the line $\mathbb{Q}x$ to precisely the three lines $\mathbb{Q}x$, $\mathbb{Q}y$, and $\mathbb{Q}(x + y)$. Define

$$E_{w,d}^{(6)} = \sum_{g \in \mathfrak{S}} gD_{w,d}, \quad E_{w,d}^{(\text{lin})} = \sum_{g \in \text{GL}_2(\mathbb{Q})} gD_{w,d}.$$

The second space is a formal enlargement. A general matrix in GL_2 need not arise from an argument change preserving the three-puncture alphabet.

Complementation changes the initial endpoint from zero to one. Its analytic realization therefore includes products with regularized endpoint constants. Whenever an analytic congruence is written below, the quotient is by the weight- w part of the product ideal generated by two positive-weight hyperlogarithms or endpoint constants, and by pure endpoint constants of weight w . No product is discarded in a stated full identity.

3.2 A formula depending only on the number of projective directions

For $w \geq 2$ and $0 \leq j \leq w$, put

$$\ell_{w,j} = \frac{1}{w} \sum_{k | \gcd(w,j)} \mu(k) \binom{w/k}{j/k}, \quad m_{w,j} = \ell_{w,j} - \ell_{w,j-1} \quad (1 \leq j \leq \lfloor w/2 \rfloor). \quad (3.2)$$

In particular, $\ell_{w,0} = 0$. The $\ell_{w,j}$ are the binary multigraded Witt numbers already used in the manuscript. The integers $m_{w,j}$ are nonnegative.

Theorem 3.1 (Finite-direction depth-orbit formula). *Let $w \geq 2$ and $1 \leq d \leq \lfloor w/2 \rfloor$. Let $\mathcal{G}$ be any nonempty finite collection of invertible rational letter maps, and suppose that $\{[gx] : g \in \mathcal{G}\}$ contains exactly k distinct points of $\mathbb{P}(V)$. Then*

$$\dim \left(\sum_{g \in \mathcal{G}} gD_{w,d} \right) = \sum_{j=1}^d m_{w,j} \min\{w - 2j + 1, k(d - j + 1)\}. \quad (3.3)$$

Consequently

$$\dim E_{w,d}^{(6)} = \sum_{j=1}^d m_{w,j} \min\{w - 2j + 1, 3(d - j + 1)\}, \quad (3.4)$$

$$\dim E_{w,d}^{(\text{lin})} = \sum_{j=1}^d m_{w,j} (w - 2j + 1). \quad (3.5)$$

Thus the finite-direction dimension depends only on k , not on the positions of the distinct rational directions.

Proof. We separate the representation-theoretic and interpolation arguments. The shuffle polynomial theorem and the standard Lyndon Lie basis imply that the weight- j subspace of Q_w , where weight here counts y letters, has dimension $\ell_{w,j}$. Equivalently, its two-variable character is the usual Witt character

$$\chi_{Q_w}(a, b) = \frac{1}{w} \sum_{k|w} \mu(k) (a^k + b^k)^{w/k}.$$

These are classical free-Lie and shuffle facts; see [4]. They may also be obtained by counting primitive binary necklaces and choosing their Lyndon representatives, as in the existing manuscript.

Extend scalars to $\mathbb{C}$ temporarily. Polynomial representations of GL_2 are completely reducible [5], and their homogeneous irreducibles of degree w have the form

$$W_{w,j} = \text{Sym}^{w-2j} V \otimes (\det V)^j.$$

This representation has one-dimensional weight spaces with y counts $j, j+1, \dots, w-j$. Comparing the character coefficients successively for $0 \leq j \leq w/2$ gives

$$Q_w \simeq \bigoplus_{j=1}^{\lfloor w/2 \rfloor} W_{w,j}^{\oplus m_{w,j}}. \quad (3.6)$$

Indeed $\ell_{w,j} = \sum_{a \leq j} m_{w,a}$, so subtraction gives exactly (3.2). This also proves the nonnegativity. Ranks of rational matrices are unchanged by the scalar extension.

Fix one copy of $W_{w,j}$, with $j \leq d$, and write $n = w - 2j$ and $r = d - j$. Its intersection with $D_{w,d}$ is

$$x^{n-r} \text{Sym}^r V \otimes (\det V)^j.$$

This is the span of $x^n, x^{n-1}y, \dots, x^{n-r}y^r$, with the determinant factor understood. A map g sends it to

$$(gx)^{n-r} \text{Sym}^r V \otimes (\det V)^j.$$

The determinant contributes a nonzero scalar and does not affect its span. Components with $j > d$ have no intersection with $D_{w,d}$.

It remains to calculate the dimension of the sum of the spaces $l_a^{n-r} \text{Sym}^r V$ for k distinct lines $[l_a]$. Associate to a linear functional λ on $\text{Sym}^n V$ the homogeneous polynomial

$$P_\lambda(s, t) = \lambda((sx + ty)^n).$$

This identifies the dual with binary forms of degree n , since every binomial coefficient involved is nonzero. The functional annihilates $l_a^{n-r} \text{Sym}^r V$ precisely when P_λ has a zero of multiplicity at least $r+1$ at the projective point $[l]$: differentiate $(l + \varepsilon m)^n$ through order r for any independent m .

For k distinct points, these conditions mean divisibility by the product of their k distinct linear vanishing forms, each to the power $r + 1$. The annihilator therefore has dimension

$$\max\{0, n - k(r + 1) + 1\}.$$

Subtracting from $n + 1$ gives $\min\{n + 1, k(r + 1)\}$. This is a direct homogeneous interpolation proof; no generic-position assumption beyond distinctness of the points is needed.

Summing independently over the copies in (3.6) proves (3.3). The six matrices have three directions, which proves (3.4). Finally, sufficiently many distinct rational directions fill each of the finitely many symmetric-power components, proving (3.5) for $\mathrm{GL}_2(\mathbb{Q})$ itself. $\square$

3.3 Consequences for the depth-two search

Since $m_{w,1} = 1$ and $m_{w,2} = \lfloor (w - 1)/2 \rfloor - 1$, the double case simplifies to

$$\dim E_{w,2}^{(6)} = \min\{w - 1, 6\} + \left(\left\lfloor \frac{w - 1}{2} \right\rfloor - 1 \right) \min\{w - 3, 3\}, \quad (3.7)$$

$$\dim E_{w,2}^{(\mathrm{lin})} = (w - 3) \left\lfloor \frac{w - 1}{2} \right\rfloor + 2, \quad w \geq 4. \quad (3.8)$$

In particular, the six-argument rank is $3\lfloor (w - 1)/2 \rfloor + 3$ for $w \geq 7$.

w	$\dim Q_w$	$\dim E_{w,2}^{(6)}$	$\dim E_{w,2}^{(\mathrm{lin})}$	$\dim(Q_w/E_{w,2}^{(6)})$
4	3	3	3	0
5	6	6	6	0
6	9	8	8	1
7	18	12	14	6
8	30	12	17	18
9	56	15	26	41
10	99	15	30	84
11	186	18	42	168
12	335	18	47	317

This table makes two different limitations visible. Components with $j > d$ cannot be reached by any linear-letter orbit of the depth- d subspace. Within components with $j \leq d$, a finite set of argument directions can still fail to span the whole symmetric power. The weight-six calculation sees only the first limitation. Both are already present at weight seven.

For fixed d and $w \geq 3d + 1$, every minimum in (3.4) selects its second entry, and telescoping gives

$$\dim E_{w,d}^{(6)} = 3 \sum_{j=1}^d \ell_{w,j}. \quad (3.9)$$

Hence, as $w \rightarrow \infty$ with d fixed,

$$\dim E_{w,d}^{(6)} \sim \frac{3w^{d-1}}{d!}, \quad \dim E_{w,d}^{(\mathrm{lin})} \sim \frac{w^d}{d!}, \quad \dim Q_w \sim \frac{2^w}{w}.$$

The estimates follow directly from the $k = 1$ term in (3.2); all other divisor terms have smaller polynomial degree for fixed j , and the usual binary Witt sum gives the last estimate. Thus a fixed depth budget and a fixed set of standard arguments span a polynomially growing subspace of an exponentially growing formal space.

3.4 A sharp height-one threshold

The general dimensions hide a particularly simple invariant family. Write

$$T_{w,h} = [x^{w-h}y^h], \quad 1 \leq h \leq w-1.$$

Its analytic representative is $\text{Li}_{w-h+1, \{1\}^{h-1}}(z)$, or $S_{w-h,h}(z)$ in Nielsen notation. The uniform bound $\lceil (w-1)/3 \rceil = \lfloor (w+1)/3 \rfloor$ is [3, Theorem 7]. The following individual-word formulation records exact membership in our specified formal presentation, using the same polynomial structure; it is not a claim of a new analytic Nielsen depth bound.

Theorem 3.2 (Exact formal depth of height-one words). *The smallest integer $d \geq 1$ for which $T_{w,h} \in E_{w,d}^{(6)}$ is*

$$d_{\min}(w, h) = \min \left\{ h, w-h, \left\lceil \frac{w-1}{3} \right\rceil \right\}. \quad (3.10)$$

In particular, $[x^{w-d-1}y^{d+1}] \notin E_{w,d}^{(6)}$ whenever $w \geq 3d+2$. The threshold is sharp: at $w = 3d+1$ every height-one word lies in $E_{w,d}^{(6)}$.

Proof. Let $L = L(V)$ be the free Lie algebra inside the tensor algebra, and put $L' = [L, L]$, $L'' = [L', L']$. Coefficient pairing identifies degree- w Lie polynomials with the dual of Q_w : Lie polynomials annihilate shuffle products, and the standard Lyndon Lie basis has the required dimension. For a letter matrix g and a Lie polynomial P , the coefficient pairing satisfies

$$\langle P, gq \rangle = \langle g^T P, q \rangle,$$

where g^T denotes the induced substitution by the transpose letter matrix. This is an adjoint identity; the contragredient group action on the dual would instead use g^{-T} . In particular, stability of a Lie subspace under all letter substitutions makes its annihilator in Q_w stable as well.

Every nonzero monomial in an element of L' contains both letters. Consequently no concatenation of two such monomials is an ordered word $x^a y^b$. Thus $T_{w,h}$ annihilates L'' . The degree- w part of L'/L'' is naturally

$$(L'/L'')_w \simeq \text{Sym}^{w-2} V \otimes \det V. \quad (3.11)$$

For completeness, ad_x and ad_y commute on L'/L'' , since their commutator is $\text{ad}_{[x,y]}$, which vanishes there. Jacobi's identity therefore spans the quotient by

$$\text{ad}_x^a \text{ad}_y^b [x, y], \quad a + b = w - 2.$$

These $w-1$ elements are independent. Map x and y to upper triangular matrices over $\mathbb{Q}[s, t]$ by

$$x \mapsto \begin{pmatrix} s & 0 \\ 0 & 0 \end{pmatrix}, \quad y \mapsto \begin{pmatrix} t & 1 \\ 0 & 0 \end{pmatrix}.$$

The displayed brackets have upper-right entries $s^{a+1}t^b$, which are distinct monomials. Their derived ideal is abelian, so this map factors through the metabelian quotient. Naturality of $[x, y]$ under letter changes proves (3.11).

It follows that every $T_{w,h}$ lies in the unique $j = 1$ summand of (3.6). Put $n = w - 2$. In that summand the depth- d orbit under the six matrices is

$$x^{n-d+1} \text{Sym}^{d-1} V + y^{n-d+1} \text{Sym}^{d-1} V + (x+y)^{n-d+1} \text{Sym}^{d-1} V,$$

with its determinant factor suppressed. By the interpolation proof, its annihilator is zero if $n < 3d$, and otherwise is

$$\Delta(s, t)^d \text{Sym}^{n-3d}(\mathbb{Q}^2)^*, \quad \Delta(s, t) = st(s - t). \quad (3.12)$$

The vector $T_{w,h}$ is a nonzero vector in its one-dimensional y -weight- h space. The corresponding coefficient in the dual binary form is its t^{h-1} coefficient, up to a nonzero scalar. To see the normalization explicitly, set $A = \text{ad}_x^{w-1} y$ and let δ be the derivation $\delta(x) = y$, $\delta(y) = 0$. For $0 \leq r \leq n$, the Lie polynomial $\delta^r A / (n)_r$ corresponds to $s^{n-r} t^r$; here $(n)_r = n(n-1) \cdots (n-r+1)$ and $(n)_0 = 1$. In this Lie realization, annihilating $D_{w,d}$ means divisibility by t^d . Applying the adjoints of R and M requires also s^d and $(s-t)^d$, respectively. This recovers (3.12) and fixes its sign convention. The coefficient of $\delta^r A / (n)_r$ at $x^{w-h} y^h$ is zero unless $r = h-1$, when it equals $(-1)^r$. Indeed

$$r! \sum_{k=0}^r (-1)^k \binom{w-1}{k} = (-1)^r r! \binom{w-2}{r} = (-1)^r (n)_r.$$

If $n \geq 3d$, the possible t exponents with nonzero coefficient in (3.12) are precisely $d, d+1, \dots, n-d$. Every such exponent occurs: multiply Δ^d by a suitable monomial of degree $n-3d$, and use the nonzero binomial coefficients in $(s-t)^d$. Thus the annihilator vanishes on $T_{w,h}$ exactly when

$$h \leq d, \quad \text{or} \quad h \geq w-d, \quad \text{or} \quad w \leq 3d+1.$$

Solving these three alternatives for the smallest d gives (3.10). $\square$

The formula combines the original and complementary depth bounds with the established uniform Nielsen bound. Our proof also tests exact nonmembership against all binary words of the permitted depth, rather than just against the height-one subfamily. At weight four it gives depth one for every height-one word; at weights six and seven it gives depth two for the middle height-one words.

3.5 Explicit primitive obstructions in every weight

The preceding proof gives actual rational certificates of nonmembership. For $w \geq 3d+2$, set $n = w-2$, $A = \text{ad}_x^{w-1} y$, and

$$\Omega_{w,d} = \sum_{r=0}^d (-1)^r \binom{d}{r} \frac{\delta^{d+r} A}{(n)_{d+r}}. \quad (3.13)$$

It corresponds to the dual form $s^{n-3d} \Delta(s, t)^d$, so it annihilates $E_{w,d}^{(6)}$. Being a Lie polynomial, it also annihilates every shuffle product of two nonempty words. Its coefficient at $x^{w-d-1} y^{d+1}$ is $(-1)^d$, proving that the target class survives.

The weight-eight double certificate is especially short:

$$\Omega_{8,2} = \frac{\delta^4 - 6\delta^3 + 12\delta^2}{360} \text{ad}_x^7 y, \quad \langle \Omega_{8,2}, x^5 y^3 \rangle = 1. \quad (3.14)$$

Hence $\text{Li}_{6,1,1}(z)$ cannot be reduced in the formal six-argument double presentation. This is an explicit nonzero functional. The corresponding Nielsen obstruction for $S_{5,3}$ is already discussed in [3, §9, preconsiderations]; our contribution here is the normalized Lie certificate with its exact full-binary verification, not priority for the nonreduction phenomenon.

3.6 An exact product completion of the known weight-seven identity

Write $F_{a,b}(z) = \text{Li}_{a,b}(z, 1)$ and $F_{a,b,c}(z) = \text{Li}_{a,b,c}(z, 1, 1)$. Put $u = 1 - z$ and $v = z/(z - 1)$.

Proposition 3.3 (Charlton–Gangl–Radchenko, Proposition 28). *Modulo positive-weight products and endpoint constants,*

$$\begin{aligned} F_{5,1,1}(z) \equiv & -3 \text{Li}_7(z) + 2F_{6,1}(z) + 2 \text{Li}_7(u) - F_{6,1}(u) \\ & -3 \text{Li}_7(v) + F_{6,1}(v). \end{aligned} \quad (3.15)$$

This is [3, Proposition 28] in our decreasing-index notation. A complete product lift is given by (3.19) below.

To state the full identity without hiding a long product polynomial, let $a = x^6y$, $b = x^5y^2$, $t = x^4y^3$, and define the explicit weight-seven word polynomial

$$\mathcal{R} = t + 3a - 2b - 2R(a) + R(b) + 3M(a) - M(b). \quad (3.16)$$

Expand it as $\mathcal{R} = \sum_w c_w w$. For this particular word polynomial define

$$\mathcal{P}_{\mathcal{R}}(z) = \sum_w c_w \sum_{k=2}^7 \frac{(-1)^k}{k} \sum_{\substack{w=w_1 \cdots w_k \\ |w_i| > 0}} \prod_{i=1}^k H_{w_i}(z). \quad (3.17)$$

All factors have strictly smaller weight than seven. This is a finite, explicit expression; the accompanying JSON lists its 864 nonzero commutatively collected product monomials. Its exact shuffle re-expansion is $\mathcal{R}$.

Put $\alpha = -\log(1 - z)$ and

$$\begin{aligned} C_7(z) &= \zeta(7) + \sum_{k=1}^5 \frac{(-1)^k \alpha^k}{k!} \zeta(7 - k), \\ C_{6,1}(z) &= \zeta(6, 1) + \sum_{k=1}^4 \frac{(-1)^k \alpha^k}{k!} \zeta(6 - k, 1). \end{aligned} \quad (3.18)$$

On the principal domain

$$\mathcal{U} = \mathbb{C} \setminus ((-\infty, 0] \cup [1, \infty)),$$

the full identity is

$$\begin{aligned} F_{5,1,1}(z) = & -3 \text{Li}_7(z) + 2F_{6,1}(z) + 2 \text{Li}_7(u) - F_{6,1}(u) \\ & -3 \text{Li}_7(v) + F_{6,1}(v) + \mathcal{P}_{\mathcal{R}}(z) - 2C_7(z) + C_{6,1}(z). \end{aligned} \quad (3.19)$$

In particular, it may be specialized at $z = i$. The factors with trailing x in (3.17) have their ordinary shuffle normalization in $\log z$, so no new undefined functions occur.

Proof. There are eighteen weight-seven binary Lyndon words. Pairing (3.16) with their eighteen standard Lie brackets gives zero in every coordinate. This exact integer calculation is reconstructed by the supplied verifier. Since these brackets form the dual of Q_7 , it proves $\mathcal{R} \in J^2$.

We explain why (3.17) is a complete product correction, rather than just a certificate in a quotient. In a connected commutative graded Hopf algebra, the convolution logarithm of the identity is

$$e = \log^*(\text{id}) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} (\text{id} - \eta\varepsilon)^{*k}.$$

It annihilates products of positive-degree elements. One elementary proof uses the convolution powers id^{*m} for nonnegative integers m : they are algebra homomorphisms because the target is commutative. In every bounded degree these powers are polynomial in m . Their homomorphism identity therefore holds for a formal parameter; taking its coefficient of degree one in that parameter gives the infinitesimal-character identity for e .

The coproduct of the shuffle Hopf algebra is deconcatenation. Thus $e(\mathcal{R}) = 0$ is exactly the formal identity saying that the shuffle re-expansion of (3.17) equals $\mathcal{R}$. Evaluation gives $H_{\mathcal{R}}(z) = \mathcal{P}_{\mathcal{R}}(z)$. Independently of this general argument, the verifier expands all 864 products and checks that their exact rational residual is zero.

For $0 < z < 1$, Landen pullback gives $H_{M(a)}(z) = \text{Li}_7(v)$ and $H_{M(b)}(z) = F_{6,1}(v)$. Path composition at the regularized endpoint one gives

$$H_{R(a)}(z) = \text{Li}_7(1 - z) - C_7(z), \quad H_{R(b)}(z) = F_{6,1}(1 - z) - C_{6,1}(z).$$

For example, the suffixes of $a = x^6y$ contribute $\zeta(7 - k)$; the complementary prefix x^k becomes $(-y)^k$ and contributes $(-1)^k \alpha^k / k!$. The suffix y has regularized endpoint value zero. The same calculation for $b = x^5y^2$ gives $\zeta(6 - k, 1)$; the suffixes y and y^2 again have zero regularized constant. These observations give exactly (3.18). Substituting the four identities into $H_{\mathcal{R}} = \mathcal{P}_{\mathcal{R}}$ proves (3.19) on $(0, 1)$. Analytic continuation proves it on $\mathcal{U}$. Discarding precisely the named product and endpoint terms gives (3.15). $\square$

3.7 Independent exact verification and remaining questions

The program `verify_depth_orbits.py` was written independently of the previous weight-six checker. It constructs binary Lyndon Lie polynomials by recursive commutators, applies the displayed integer letter substitutions, and computes matrix ranks over $\mathbb{Q}$ using integer elimination with gcd normalization. It imports no numerical polylogarithm evaluator and no previous relation matrix.

At weights 2 through 10, it checks all 25 pairs (w, d) with $1 \leq d \leq \lfloor w/2 \rfloor$, both for the six maps and for enough distinct rational linear directions to fill the full linear orbit. It additionally checks 34 depth-two cases involving one through five irregularly spaced rational directions. Every rank agrees with (3.3). A further 155 exact membership checks verify the height-one threshold for every relevant height and depth through weight ten. At weight eight the explicit obstruction is checked on all 222 transformed binary words of depth at most two, including trailing-zero words, and on all 1792 nonempty binary shuffle products. Its target pairing is exactly one. The weight-seven product certificate is independently re-expanded term by term with exact fractions.

These finite checks audit the conventions and examples. The all-weight statements follow from the representation and interpolation proofs, without extrapolation from a table. The Witt character, complete reducibility, binary-form interpolation, and convolution logarithm are classical ingredients. The finite-direction formula treats the full binary quotient; the height-one component recovers established Nielsen structure. The independently constructed exact certificates and product lift make these statements directly auditable. None of the height-one reductions is presented as a new identity, and no claim of worldwide priority is made for the general rank formula.

Several next questions are now sharply posed. First, one can construct compact product corrections for other height-one reductions predicted by (3.10), replacing the deliberately uniform all-cuts expression by smaller identities. Second, the irreducible components with $j > d$ isolate the exact missing representations for future argument alphabets. Third, it remains to determine which of the formal obstructions survive after specialization to Gaussian or Eisenstein points and after

adding their full colored relations. Finally, the conjectured short S_6 identity remains a separate mixed-color problem: none of the binary rank or height-one theorems proves or refutes it.

4 An isolated sixth-order radial degeneracy and two radial extrema

The existing fractional-turning analysis proves that the quadratic radial coefficient has a unique zero in the outer order, and exhibits both signs of the quartic coefficient along that threshold. Its final numerical diagnostic suggests a simultaneous zero of those two coefficients near $(a, b) = (1.00143, 0.99749)$. We now certify that point, prove that it is locally isolated, determine its sixth-order coefficient, and derive its local radial geometry. In particular, some nearby fractional-order curves have a strict local minimum followed by a strict local maximum. This is a stronger failure of radial monotonicity than the previously proved single-extremum branches.

The uniqueness asserted below is local to a stated rational rectangle. No assertion about the total number of quartic transition points on the full interval $0 < b < 1$ is needed or implied.

4.1 A convergent all-orders equation for the radial coefficients

For $a, b > 0$, write

$$p_n = p_n(a, b) = \frac{H_{n-1}^{(b)}}{n^a}, \quad F_{a,b}(z) = \sum_{n \geq 2} p_n z^n.$$

Let $\theta_{a,b}(\rho)$ denote the unique zero of $\Im F_{a,b}(\rho e^{i\theta})$ in the upper open semicircle, as established by the manuscript's angular-zero theorem, and set

$$\eta_{a,b}(\rho) = \frac{\cos \theta_{a,b}(\rho)}{\rho}, \quad t = \rho^2.$$

Only the local branch at $\rho = 0$ is needed for the constructions below. It can also be obtained directly by the implicit-function argument in the following lemma.

Lemma 4.1 (Binomial equation for every radial Taylor coefficient). *Define, for $j \geq 0$,*

$$B_j(v) = \sum_{k=0}^{j+1} (-1)^{j+1-k} \binom{j+1}{k} (2v)^k p_{2j+3-k}. \quad (4.1)$$

Near $t = 0$, the radial zero is the analytic solution of

$$E(a, b, t, v) := \sum_{j \geq 0} t^j B_j(v) = 0, \quad v(0) = \mu := \frac{p_3}{2p_2}. \quad (4.2)$$

The solution is jointly real analytic in (a, b, t) near every point with $a, b > 0$ and $t = 0$. Consequently

$$\eta_{a,b}(\rho) = \mu + K\rho^2 + Q\rho^4 + R\rho^6 + O(\rho^8), \quad (4.3)$$

where

$$K = -\frac{B_1(\mu)}{2p_2}, \quad (4.4)$$

$$Q = -\frac{B'_1(\mu)K + B_2(\mu)}{2p_2}, \quad (4.5)$$

$$R = -\frac{B'_1(\mu)Q + 4p_3K^2 + B'_2(\mu)K + B_3(\mu)}{2p_2}. \quad (4.6)$$

In particular, at a simultaneous zero of K and Q ,

$$R = -\frac{16p_5\mu^4 - 32p_6\mu^3 + 24p_7\mu^2 - 8p_8\mu + p_9}{2p_2}. \quad (4.7)$$

Proof. Use $\sin(n\theta) = \sin\theta U_{n-1}(\cos\theta)$ and the standard finite Chebyshev expansion [6]

$$U_{n-1}(c) = \sum_{\ell=0}^{\lfloor (n-1)/2 \rfloor} (-1)^\ell \binom{n-1-\ell}{\ell} (2c)^{n-1-2\ell}.$$

Put $\cos\theta = \rho v$ and divide the imaginary-part equation by $\rho^3 \sin\theta$. A monomial indexed by (n, ℓ) has radial power $\rho^{2(n-2-\ell)}$. For fixed $j = n - 2 - \ell$, write $k = 2j + 3 - n$. Then $0 \leq k \leq j + 1$, and the resulting coefficient is exactly (4.1).

The rearrangement gives a convergent analytic series, not only a formal one. For real positive a, b , one may use $0 < p_n \leq n$. On a compact complex neighborhood of a fixed positive parameter pair the same coefficients have a uniform bound $|p_n| \leq Cn^M$ for some finite M . For $|v| \leq V$ this bounds $|B_j(v)|$ by $C(2j+3)^M(1+2V)^{j+1}$. Thus the series in (4.2) converges normally when $|t|(1+2V) < 1$.

Since $B_0(v) = 2p_2v - p_3$, its v derivative is $2p_2 > 0$. The analytic implicit-function theorem applies and gives the stated joint analyticity. Expanding E to degree three in t proves (4.4)–(4.6), because

$$\begin{aligned} B_1(v) &= 4p_3v^2 - 4p_4v + p_5, \\ B_2(v) &= 8p_4v^3 - 12p_5v^2 + 6p_6v - p_7, \\ B_3(v) &= 16p_5v^4 - 32p_6v^3 + 24p_7v^2 - 8p_8v + p_9, \end{aligned}$$

and $B_1'' = 8p_3$. Setting $K = Q = 0$ gives (4.7). $\square$

There is also a useful signed-moment interpretation. Whenever $p_n = \int_0^1 u^{n-1} d\nu(u)$ with $\nu([0, 1]) = 0$, the finite polynomial identity (4.1) becomes

$$B_j(v) = \int_0^1 [u(2v - u)]^{j+1} d\nu(u). \quad (4.8)$$

Consequently a vanishing quadratic and quartic radial coefficient means that the first three moments of the quadratic expression $u(2\mu - u)$ vanish against this signed measure. The fourth such moment determines the sixth radial coefficient by (4.7). The finite-measure hypothesis holds at all parameter pairs used below.

4.2 Exact isolation of the degeneracy

All decimal endpoints in the next theorem denote terminating decimals, hence exact rational numbers.

Theorem 4.2 (Certified simultaneous zero and negative sixth coefficient). *There is exactly one pair $(a_{\dagger}, b_{\dagger})$ in the open rectangle*

$$1.00143018096149513812 < a < 1.00143018096149513814, \quad (4.9)$$

$$0.99749378987342042107 < b < 0.99749378987342042108 \quad (4.10)$$

such that $K(a, b) = Q(a, b) = 0$. At that point, and indeed throughout the closed rectangle, the following rigorous bounds hold:

$$-3.368745 \cdot 10^{-11} < R < -3.368744 \cdot 10^{-11}, \quad (4.11)$$

$$4.87028 \cdot 10^{-8} < Q_b - Q_a K_b / K_a < 4.87029 \cdot 10^{-8}, \quad (4.12)$$

$$-8.31515 \cdot 10^{-10} < K_a Q_b - K_b Q_a < -8.31513 \cdot 10^{-10}. \quad (4.13)$$

Moreover $K_a < 0$, $K_b < 0$, and $Q_a < 0$ there. Thus the map $(a, b) \mapsto (K, Q)$ is an analytic local coordinate system at $(a_\dagger, b_\dagger)$, and the quartic coefficient crosses zero strictly increasingly along the local branch $K = 0$ as b increases.

Proof. The accompanying verifier `certify_radial_degeneracy.py` uses only Python's standard-library integer and rational arithmetic. Its output, `radial_degeneracy_certificate.json`, records every interval as exact rational endpoints, with a separate outward decimal display. The finite computation and its analytic implications are as follows.

Let b_- and b_+ be the lower and upper endpoints in (4.10). The verifier encloses the corresponding zeros by $L_- < a_*(b_-) < U_-$ and $L_+ < a_*(b_+) < U_+$, where

$$L_- = 1.00143018096149513813198112513591610363,$$

$$U_- = 1.00143018096149513813198112513591610365,$$

$$L_+ = 1.00143018096149513812627198907246165442,$$

$$U_+ = 1.00143018096149513812627198907246165445.$$

Specifically, K is strictly positive at each lower a endpoint and strictly negative at each upper a endpoint. The entire first thin interval has $Q < 0$, while the entire second has $Q > 0$.

For transparency, the two quartic enclosures imply the simpler rational inequalities

$$-2.244 \cdot 10^{-28} < Q(a_*(b_-), b_-) < -2.242 \cdot 10^{-28},$$

$$2.626 \cdot 10^{-28} < Q(a_*(b_+), b_+) < 2.629 \cdot 10^{-28}.$$

Across the full closed rectangle, the verifier also proves

$$-0.017073210967 < K_a < -0.017073210965,$$

$$-0.009747328446 < K_b < -0.009747328443,$$

$$-0.002284745315 < Q_a < -0.002284745313,$$

as well as (4.11)–(4.13).

Here are the elementary bounds underlying all transcendental enclosures. For each integer $1 \leq n \leq 9$, put $z = (n - 1)/(n + 1)$ and use the first 360 terms of

$$\log n = 2 \sum_{j \geq 0} \frac{z^{2j+1}}{2j+1}.$$

The positive omitted tail is at most $2z^{721}/(721(1 - z^2))$. For every nonnegative rational argument x used in the computation, $x < 3$. The Taylor polynomial for e^x through degree 100 has positive remainder at most

$$\frac{x^{101}}{101!} \frac{1}{1 - x/102}.$$

Exponentials at interval endpoints, followed by reciprocation, enclose all powers n^{-a} and n^{-b} . Each arithmetic operation is rounded outward to integer multiples of 2^{-192} . First derivatives are enclosed by the ordinary product and quotient rules, starting from

$$\partial_a p_n = -(\log n)p_n, \quad \partial_b p_n = -n^{-a} \sum_{m < n} (\log m)m^{-b}.$$

Substitution in the finite formulas of Lemma 4.1 therefore proves all recorded signs by finitely many rational inequalities. No floating-point sign decision enters the certificate.

It remains to explain why these signs prove isolation. Since $K_a < 0$ and $K_b < 0$ throughout the rectangle, the endpoint brackets imply that K is positive on its lower a edge and negative on its upper a edge. For every $b \in [b_-, b_+]$ there is consequently a unique zero $a_*(b)$ between those edges. The implicit-function theorem makes this branch analytic and gives

$$a'_*(b) = -K_b/K_a.$$

The quartic coefficient on it has derivative $Q_b - Q_a K_b/K_a > 0$ by (4.12). Its opposite endpoint signs give exactly one zero in (b_-, b_+) , and hence exactly one simultaneous zero in the rectangle. Finally (4.13) gives the local inverse coordinate map, while (4.11) gives the negative sixth coefficient. $\square$

Remark 4.3 (What the certificate settles). The original decimal location is upgraded to a proved rational enclosure with local uniqueness, transversal quartic crossing, and a nonzero sixth coefficient. The proof does not establish uniqueness of the transition among all $0 < b < 1$. Nor does a local coefficient sign prove monotonicity at every radius. The next results derive exactly the local consequences warranted by these certified data.

4.3 A quarter-power branch of strict maxima

Set

$$R_{\dagger} = R(a_{\dagger}, b_{\dagger}), \quad r = -R_{\dagger} > 0, \quad \mu_{\dagger} = \frac{p_3(a_{\dagger}, b_{\dagger})}{2p_2(a_{\dagger}, b_{\dagger})}.$$

Theorem 4.4 (Sixth-order decrease and quarter-power maximum birth). *At the certified pair,*

$$\eta_{a_{\dagger}, b_{\dagger}}(\rho) = \mu_{\dagger} - r\rho^6 + O(\rho^8),$$

so the normalized-radius curve strictly decreases for sufficiently small positive ρ . There are $\delta_0, \rho_0 > 0$ such that for every $0 < \delta < \delta_0$ the curve $\eta_{a_{\dagger}-\delta, b_{\dagger}}$ has exactly one critical point in $(0, \rho_0)$, a strict local maximum. Its radius is

$$\rho_{\max}(\delta) = \left(\frac{K_a(a_{\dagger}, b_{\dagger})}{3R_{\dagger}} \right)^{1/4} \delta^{1/4} + O(\delta^{3/4}). \quad (4.14)$$

The positive prefactor has the certified rational bounds

$$114.006986 < \left(\frac{K_a(a_{\dagger}, b_{\dagger})}{3R_{\dagger}} \right)^{1/4} < 114.006989.$$

For $a > a_{\dagger}$ sufficiently close, at the same $b = b_{\dagger}$, there are no critical points in $(0, \rho_0)$ and the curve strictly decreases there.

Proof. Write $\Phi(a, b, t) = \eta_{a,b}(\sqrt{t})$ for the analytic solution of Lemma 4.1, and let $U = \partial_t \Phi$. At the certified pair,

$$U(a_{\dagger}, b_{\dagger}, t) = 3R_{\dagger}t^2 + O(t^3).$$

Moreover $U_{tt} = 6R_{\dagger} < 0$ at $t = 0$. Choose a small fixed parameter and t rectangle on which $U_{tt} < 0$, and then a positive upper endpoint t_0 on which $U < 0$ for all the sufficiently nearby parameters considered here.

For $a = a_{\dagger} - \delta$, the initial value is $U(a, b_{\dagger}, 0) = K(a, b_{\dagger}) > 0$. Strict concavity and the negative value at t_0 imply exactly one zero in $(0, t_0)$; at it $U_t < 0$. Thus $\partial_{\rho} \eta = 2\rho U$ changes from positive to negative, giving a strict maximum.

To find its scale, put $\delta = s^2$ and $t = sv$. Analyticity makes

$$\frac{U(a_{\dagger} - s^2, b_{\dagger}, sv)}{s^2} = -K_a(a_{\dagger}, b_{\dagger}) + 3R_{\dagger}v^2 + O(s)$$

an analytic expression through $s = 0$. Its positive zero at $s = 0$ is $v_0 = (K_a/(3R_{\dagger}))^{1/2}$ and has nonzero v derivative. The implicit-function theorem therefore gives $v(s) = v_0 + O(s)$. Taking the positive square root of $t = \sqrt{\delta} v(\sqrt{\delta})$ proves (4.14). The displayed prefactor enclosure is checked by comparing the fourth powers of its rational endpoints against the exact interval for $K_a/(3R)$; no numerical fourth root is required.

For $a > a_{\dagger}$ sufficiently close, both $K < 0$ and $Q < 0$, because $K_a < 0$ and $Q_a < 0$ at the certified pair. Hence $U(0) < 0$ and $U_t(0) = 2Q < 0$. Strict concavity forces $U < 0$ throughout the positive t interval. At the certified pair itself, the leading term $3R_{\dagger}t^2$ gives the stated strict decrease. $\square$

4.4 A proved region containing two radial extrema

Theorem 4.5 (A minimum followed by a maximum). *Use the analytic local coordinates $k = K(a, b)$ and $q = Q(a, b)$ centered at $(a_{\dagger}, b_{\dagger})$. There is an analytic function $\chi(q)$ near zero with*

$$\chi(q) = \frac{q^2}{3R_{\dagger}} + O(q^3) = -\frac{q^2}{3r} + O(q^3) \quad (4.15)$$

and a fixed $\rho_0 > 0$ such that, for sufficiently small $q > 0$:

1. If $\chi(q) < k < 0$, the normalized-radius curve has exactly two critical points in $(0, \rho_0)$. The first is a strict local minimum and the second is a strict local maximum.
2. If $k = \chi(q)$, these two critical points coalesce at a positive radius. The derivative vanishes to order two there, and the curve strictly decreases through this stationary point.
3. If $k < \chi(q)$ and (k, q) remains sufficiently near zero, the curve strictly decreases throughout $(0, \rho_0)$.

In particular there is an explicit analytic parameter path $(a(\varepsilon), b(\varepsilon))$ defined for small $\varepsilon > 0$ by

$$K(a(\varepsilon), b(\varepsilon)) = -\frac{9}{4}r\varepsilon^2, \quad Q(a(\varepsilon), b(\varepsilon)) = 3r\varepsilon. \quad (4.16)$$

Its two critical radii satisfy

$$\rho_{\min}(\varepsilon) = \sqrt{\varepsilon/2} + O(\varepsilon^{3/2}), \quad (4.17)$$

$$\rho_{\max}(\varepsilon) = \sqrt{3\varepsilon/2} + O(\varepsilon^{3/2}). \quad (4.18)$$

All these parameters remain fractional positive orders with $a > 1$ and $0 < b < 1$ after the neighborhood is made sufficiently small.

Proof. Let $(a, b) = P(k, q)$ be the analytic inverse coordinate map guaranteed by Theorem 4.2, and continue to write $\Phi(k, q, t) = \eta_{P(k, q)}(\sqrt{t})$. Its derivative has the exact analytic form

$$U(k, q, t) = k + 2qt + 3R(P(k, q))t^2 + O(t^3).$$

At the origin $U_{tt} = 6R_{\dagger} < 0$. Work in a small fixed rectangle where $U_{tt} < 0$ and with a positive endpoint t_0 where $U < 0$ for all sufficiently small (k, q) .

The equation $U_t(k, q, t) = 0$ has a unique analytic solution $t = t_m(k, q)$ near the origin, because its t derivative there is $6R_{\dagger} \neq 0$. Since $U_t(k, 0, 0) = 0$ identically, uniqueness gives $t_m(k, 0) = 0$. Expanding at the origin yields

$$t_m(k, q) = -\frac{q}{3R_{\dagger}} + O((|k| + |q|)^2).$$

Thus $t_m > 0$ for small $q > 0$; more precisely $t_m = qA(k, q)$ with A analytic and $A(0, 0) = -1/(3R_{\dagger}) > 0$.

Define the local maximum value of U by $H(k, q) = U(k, q, t_m(k, q))$. Its derivative in k at the origin is one. The implicit-function theorem gives an analytic curve $k = \chi(q)$ on which $H = 0$, with $H > 0$ for $k > \chi(q)$ and $H < 0$ for $k < \chi(q)$ nearby. Expanding H gives

$$H(k, q) = k - \frac{q^2}{3R_{\dagger}} + O((|k| + |q|)^3),$$

and therefore (4.15).

If $\chi(q) < k < 0$, then $U(0) = k < 0$, its unique interior maximum is positive, and $U(t_0) < 0$. Strict concavity gives exactly two simple positive zeros, with signs $-, +, -$ in the three intervening intervals. Since $\partial_{\rho}\eta = 2\rho U$, they are respectively a strict minimum and a strict maximum. At $k = \chi(q)$ the maximum equals zero, so the sole zero is double and U is negative on both sides. For $k < \chi(q)$ even its maximum is negative. These prove the three local classifications. At the double zero $t_m > 0$, $U = U_t = 0$ and $U_{tt} < 0$; changing to $\rho = \sqrt{t}$ preserves the multiplicity two of the first derivative.

The inverse map P makes (4.16) an analytic path. Put $t = \varepsilon v$. Along it, division by $3r\varepsilon^2$ gives an analytic expression through $\varepsilon = 0$:

$$\frac{U(-\frac{9}{4}r\varepsilon^2, 3r\varepsilon, \varepsilon v)}{3r\varepsilon^2} = -\frac{3}{4} + 2v - v^2 + O(\varepsilon).$$

The two zeros at $\varepsilon = 0$ are $v = 1/2$ and $v = 3/2$, both simple. The implicit-function theorem yields $v_-(\varepsilon) = 1/2 + O(\varepsilon)$ and $v_+(\varepsilon) = 3/2 + O(\varepsilon)$. Positive square roots give (4.17)–(4.18); concavity proves there are no additional critical points in the chosen local interval. Finally $(a_{\dagger}, b_{\dagger})$ is strictly inside $a > 1$, $0 < b < 1$, so continuity preserves these inequalities in a small neighborhood. $\square$

4.5 An explicit rational pair with exactly two small-radius extrema

The analytic parameter path above proves the mechanism. The following independent certificate supplies a concrete rational pair and fixed radial intervals. It encloses the actual implicit zero of E , including the omitted infinite tail.

Theorem 4.6 (A concrete minimum and maximum). *For the exact rational orders*

$$a = 1.0014289951689850677, \quad b = 0.9974958668828783544, \quad (4.19)$$

the normalized-radius curve has exactly two critical points in $0 < \rho < \sqrt{7/4000}$. They are a strict local minimum and a strict local maximum, respectively, with certified brackets

$$\frac{1}{\sqrt{4000}} < \rho_{\min} < \frac{1}{\sqrt{1000}} < \rho_{\max} < \sqrt{\frac{7}{4000}}. \quad (4.20)$$

The assertion does not classify critical points at larger radii.

Proof. Put $\Phi(t) = \eta_{a,b}(\sqrt{t})$ and $U(t) = \Phi'(t)$. The verifier `certify_two_extrema.py` gives the exact rational enclosures

t	certified interval for $U(t)$
1/4000	$(-3.158, -3.156) 10^{-17}$
1/1000	$(2.537, 2.538) 10^{-17}$
7/4000	$(-3.136, -3.135) 10^{-17}$.

Throughout the entire interval $0 \leq t \leq 7/4000$, it also proves

$$-2.1 \cdot 10^{-10} < \Phi'''(t) < -1.9 \cdot 10^{-10}. \quad (4.21)$$

Here are complete bounds for the infinite series and implicit calculation underlying these signs.

Sum (4.2) through $j = 17$, so the omitted tail begins at $N = 18$. On $|v| \leq 1$ and $0 \leq t \leq t_0 := 7/4000$, any mixed derivative with $r + s \leq 3$ has omitted tail bounded in absolute value by

$$T_{r,s} = 15 \left(\frac{4}{3}\right)^s t_0^{-r} \frac{(3t_0)^N N^{r+s+1} (r+s+1)!}{(1-3t_0)^{r+s+2}}. \quad (4.22)$$

To prove this, differentiate (4.1), use $p_n \leq n$ and the binomial theorem, and write $(j)_r$ for the falling factorial. For $j \geq N$ the resulting summand is at most

$$(j)_r (2j+3) 2^s (j+1)_s 3^{j+1-s} t_0^{j-r} \leq 15(4/3)^s j^{r+s+1} 3^j t_0^{j-r}.$$

For $q = 3t_0$, $d = r + s + 1$, and $j = N + \ell$, use $j^d \leq N^d (\ell + 1)^d$ and $(\ell + 1)^d \leq d! \binom{\ell + d}{d}$. Summing the binomial generating function proves (4.22). The same formula with each sample's smaller positive t gives its sharper pointwise bounds.

Every finite coefficient is enclosed using outward 192-bit rational interval arithmetic. The logarithms up to $\log 37$ are evaluated by binary range reduction: write $n = 2^k y$, $1 \leq y < 2$, and use $\log n = k \log 2 + 2 \operatorname{arctanh}((y-1)/(y+1))$. The last argument is less than $1/3$. One hundred terms of its positive series, with the geometric tail used above, suffice. Exponentials use one hundred Taylor terms and the same first-omitted-term geometric bound; all arguments are less than five. Thus the powers and every mixed derivative of the finite sum are rigorously enclosed.

The actual implicit root is trapped uniformly in the strip

$$0 \leq t \leq 7/4000, \quad 0.499999752369655 \leq v \leq 0.499999752369657.$$

Including (4.22), the verifier proves $E < 0$ on the lower v edge, $E > 0$ on the upper edge, and $E_v > 0$ throughout the strip. Hence each t has a unique root in this strip, depending analytically on t ; the angular-zero theorem identifies it with $\Phi(t)$. At each sample point, a further rational v bracket of width 10^{-34} encloses the root. These finer brackets, edge signs, and all derivative intervals are recorded in `two_extrema_certificate.json`.

Implicit differentiation gives

$$\begin{aligned}
U &= -E_t/E_v, \\
V := \Phi'' &= -\frac{E_{tt} + 2E_{tv}U + E_{vv}U^2}{E_v}, \\
\Phi''' &= -\frac{E_{ttt} + 3E_{ttv}U + 3E_{tvv}U^2 + E_{vvv}U^3 + 3E_{tv}V + 3E_{vv}UV}{E_v}.
\end{aligned}$$

Substituting the exact mixed-derivative enclosures proves both the three displayed signs and (4.21).

Thus U is strictly concave. Its signs $-, +, -$ force one zero in each of $(1/4000, 1/1000)$ and $(1/1000, 7/4000)$. Strict concavity permits no third zero and makes both zeros simple: the slope at the first is positive, while the slope at the second is negative. Since $\partial_\rho \eta = 2\rho U$, they are exactly the asserted strict minimum and maximum. Taking square roots of the t brackets gives (4.20). $\square$

4.6 Scope and further questions

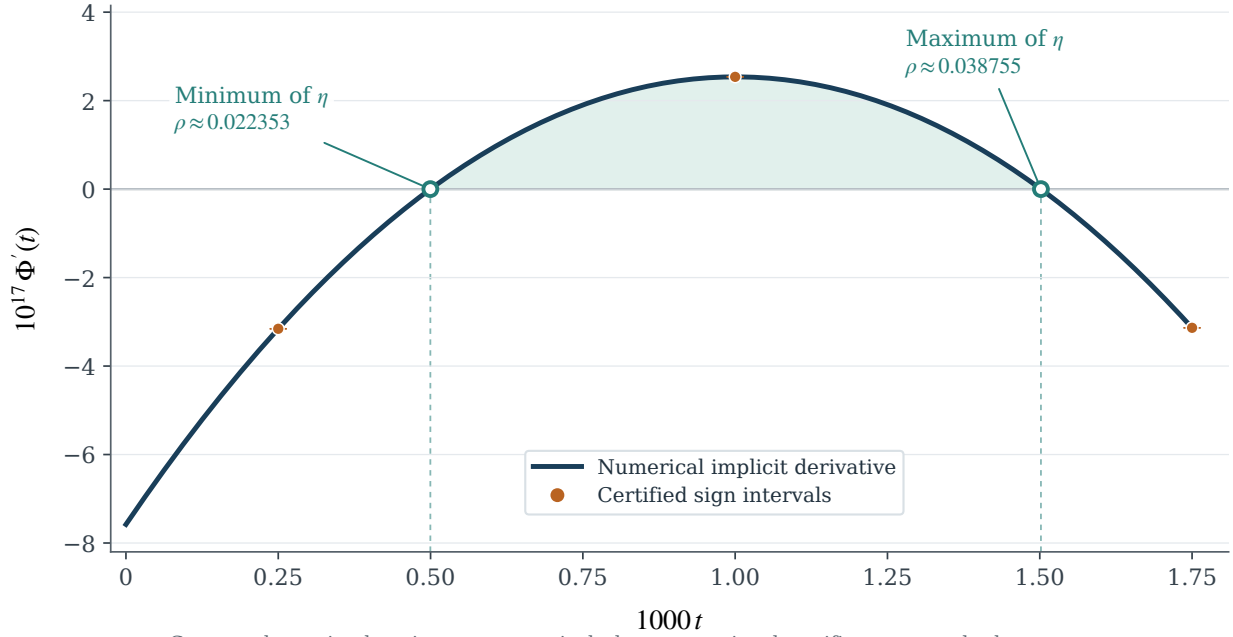
These theorems answer the existing question about the higher radial behavior at at least one quartic transition. They establish a certified local configuration, analytic parameter families, and one explicit rational pair with exactly two extrema on a fixed interval. The general bifurcation theorems do not give uniform lower bounds on their neighborhood sizes δ_0 , ρ_0 , or the allowed ε . The explicit example shows how convergent remainder bounds can make an individual instance fully quantitative.

The next concrete research questions are:

1. Is the certified $b_\dagger$ the only zero of $Q(a_*(b), b)$ in $0 < b < 1$? The local uniqueness proof leaves the rest of that interval unclassified.
2. How many radial critical points can occur on the entire interval $0 < \rho \leq 1$? The two-extremum theorem supplies a lower bound of two for the maximal possible count in the fractional family.
3. Can the integer-order all-radius monotonicity conjecture be proved from the positive double kernel or the one-change signed density? None of the present counterexamples has integer outer order.
4. Can the explicit two-extremum certificate be extended to a substantial open rational parameter rectangle, and can its critical radii be enclosed much more sharply? This would make the local bifurcation region quantitatively visible.
5. What is the analytic structure of the coefficient sequence near $(a, b) = (1, 1)$, where $\eta \equiv 1/2$ and every nonconstant radial coefficient vanishes? The nearby isolated degeneracy suggests studying that limit by a joint parameter expansion.

Two turning points of the normalized angular zero

$$a = 1.0014289951689850677, \quad b = 0.9974958668828783544, \quad t = \rho^2$$



Curve and crossing locations are numerical; the exact rational certificate proves both extrema.

Figure 2: The actual implicit derivative for the rational orders in (4.19). The blue curve and crossing locations are numerical evaluations at 85 decimal digits; the orange markers show the three exact rational sign enclosures. The independent certificate proves $\Phi''' < 0$ on the entire displayed interval, so these signs establish exactly two simple critical points. The much finer decimal crossing locations are diagnostic, not additional certified radius brackets.

5 A square-root transition in the late reflected residues

This section concerns a different simultaneous limit from the transition $n/(m+1) - \log m = O(1)$ for the moments themselves. Here the late residue index k grows, and the reflected exponent is allowed to satisfy $m = O(\sqrt{k})$. The fixed- m asymptotic in the manuscript does not cover this limit. The novelty assertion here is relative to the pinned repository: no claim of priority over all literature is made. We prove a Gaussian damping factor, give its first correction, and then obtain a sharp truncation theorem for a growing reflected exponent.

Put

$$f(x) = \log \Gamma(x), \quad B(x) = \log \Gamma(1-x), \quad M_{n,m} = \int_0^1 f(x)^n f(1-x)^m dx,$$

and retain the manuscript's residues

$$A_{k,m} = \frac{1}{k} [x^{k-1}] \Gamma(1+x)^k B(x)^m. \quad (5.1)$$

The parameters k, m are integers with $k \geq 1$, $m \geq 0$. The coefficients vanish when $k \leq m$. In the meromorphic transform $F_m(z) = \int_0^1 \Gamma(x)^z B(x)^m dx$, the actual residue is $\text{Res}_{z=k} F_m(z) = -k A_{k,m}$, so its sign is the opposite of the coefficient sign. The standard Taylor series for log gamma used below follows from DLMF, Section 5.7[7]. Lagrange inversion and the underlying saddle and singularity methods are classical[8, 9]; the moving-parameter estimates are proved explicitly here. Define

$$\phi(u) = \Gamma(1-u), \quad a(u) = -\log \Gamma(1+u), \quad D = u \frac{d}{du}.$$

Let $\tau \in (0, 1)$ be the unique solution of $D \log \phi(\tau) = 1$, and put

$$\rho = \frac{\tau}{\phi(\tau)}, \quad \Lambda = -\log \rho, \quad b = a(\tau), \quad v = D^2 \log \phi(\tau), \quad C = \frac{\tau}{\sqrt{2\pi v}}. \quad (5.2)$$

All these constants are positive. Use a local real branch of $\log a$ at τ and set

$$r_j = D^j \log a(\tau), \quad \kappa_j = D^j \log \phi(\tau), \quad \delta_\Gamma = \frac{r_1^2}{2v}. \quad (5.3)$$

Thus $\kappa_2 = v$. The subscript on δ_Γ distinguishes this constant from a small contour neighborhood. The damping constant is strictly positive. Indeed $D \log \phi(1/2) = \gamma/2 + \log 2 < 1$ and $\psi(3/2) = 2 - \gamma - 2 \log 2 > 0$, using $\gamma < 3/5$ and $\log 2 < 7/10$. Hence $\tau > 1/2$, $\psi(1+\tau) > 0$, and $r_1 = -\tau \psi(1+\tau)/b < 0$. Numerically,

$$b = 0.1206254661326052\dots, \quad r_1 = -0.1684096130814827\dots, \quad \delta_\Gamma = 0.00624269203454726\dots$$

These decimals only illustrate the constants; their defining equations are used in the proof.

Theorem 5.1 (Uniform late-residue transition). *Fix $L < \infty$. Uniformly for integers $0 \leq m \leq L\sqrt{k}$ as $k \rightarrow \infty$, with $\xi = m/\sqrt{k}$,*

$$A_{k,m} = (-1)^{k+m-1} C b^m \rho^{-k} k^{-3/2} e^{-\delta_\Gamma \xi^2} \left(1 + \frac{P_1(\xi)}{\sqrt{k}} + O_L(k^{-1}) \right), \quad (5.4)$$

where

$$\begin{aligned} P_1(\xi) = & \left[-\frac{r_1 + r_2/2}{v} + \frac{\kappa_3 r_1}{2v^2} \right] \xi \\ & + \left[\frac{r_2 r_1^2}{2v^2} - \frac{\kappa_3 r_1^3}{6v^3} \right] \xi^3. \end{aligned} \quad (5.5)$$

For every fixed $J \geq 0$ there are explicitly computable real polynomials $P_0, \dots, P_J$, with $P_0 = 1$, such that the parenthesis in (5.4) equals

$$\sum_{j=0}^J k^{-j/2} P_j(\xi) + O_{L,J}(k^{-(J+1)/2}).$$

In particular the coefficient signs are uniformly $(-1)^{k+m-1}$ for all sufficiently large k in this moving range. If $m/\sqrt{k} \rightarrow \xi_0$, then the ratio of $A_{k,m}$ to its fixed- m leading expression converges to $e^{-\delta_r \xi_0^2}$.

Proof. Replacing x by $-u$ in (5.1) gives

$$A_{k,m} = \frac{(-1)^{k+m-1}}{k} [u^{k-1}] \phi(u)^k a(u)^m.$$

The series of ϕ has strictly positive coefficients, because

$$\log \phi(u) = \gamma u + \sum_{j \geq 2} \frac{\zeta(j)}{j} u^j.$$

Consequently

$$Q(t) = \frac{\phi(\tau e^{it})}{\phi(\tau)} e^{-it}$$

satisfies $|Q(t)| < 1$ for $0 < |t| \leq \pi$, and

$$\log Q(t) = -vt^2/2 - i\kappa_3 t^3/6 + \kappa_4 t^4/24 + O(t^5).$$

Cauchy's coefficient formula is

$$A_{k,m} = \frac{(-1)^{k+m-1} \tau \rho^{-k} b^m}{2\pi k} \int_{-\pi}^{\pi} Q(t)^k e^{it} \left(\frac{a(\tau e^{it})}{b} \right)^m dt. \quad (5.6)$$

No logarithm of a on the full circle is required: a^m is an analytic integer power. Choose a small fixed $\epsilon > 0$ such that a is nonzero near the arc $|t| \leq \epsilon$. Outside that arc, compactness gives a bound $O(q^k H^m)$ with $q < 1$ and a fixed H . Since $m \leq L\sqrt{k}$, this is exponentially small in k .

On the small arc use the logarithm with value $\log b$ at zero. Real Taylor coefficients imply

$$\log \frac{a(\tau e^{it})}{b} = ir_1 t - r_2 t^2/2 - ir_3 t^3/6 + O(t^4).$$

Its real part is $O(t^2)$. After reducing ϵ if necessary, $|Q(t)|^k |a(\tau e^{it})/b|^m \leq e^{-ckt^2}$ with $c > 0$, uniformly in the stated range once k is large. Thus the portion $k^{-2/5} \leq |t| \leq \epsilon$ is $O(e^{-ck^{1/5}})$. Set $t = z/\sqrt{k}$ on the remaining arc. Its integrand is

$$e^{-vz^2/2 + ir_1 \xi z} \left[1 + \frac{1}{\sqrt{k}} \left(iz - \frac{i\kappa_3 z^3}{6} - \frac{\xi r_2 z^2}{2} \right) + O_L(k^{-1}(1 + |z|^6)) \right].$$

The error is bounded by an integrable polynomial times a fixed Gaussian; this follows from the analytic Taylor remainder and the preceding Gaussian bound. Extending the z interval to $\mathbb{R}$ has an exponentially small cost. The leading integral is

$$\int_{\mathbb{R}} e^{-vz^2/2 + ir_1 \xi z} dz = \sqrt{\frac{2\pi}{v}} e^{-r_1^2 \xi^2/(2v)}.$$

It is bounded away from zero relative to $\sqrt{2\pi/v}$ for $0 \leq \xi \leq L$, so all subsequent error estimates remain relative and uniform.

Gaussian differentiation gives the normalized moments

$$\langle z \rangle = \frac{ir_1\xi}{v}, \quad \langle z^2 \rangle = \frac{1}{v} - \frac{r_1^2\xi^2}{v^2}, \quad \langle z^3 \rangle = i \left(\frac{3r_1\xi}{v^2} - \frac{r_1^3\xi^3}{v^3} \right).$$

Substituting these into the term of order $k^{-1/2}$ gives (5.5). For any higher fixed order, expand the two analytic logarithms and e^{it} to the corresponding finite order, then integrate the resulting polynomial against this same complex Gaussian. Gaussian domination bounds the remainder. Differentiation of the Gaussian Fourier transform shows that every normalized coefficient is a polynomial in ξ . This proves the full expansion and all assertions of the theorem. $\square$

A completely explicit coefficient prescription is available. With $\varepsilon = k^{-1/2}$, expand

$$\exp \left\{ i\varepsilon z + \sum_{j \geq 3} \frac{\kappa_j (iz)^j}{j!} \varepsilon^{j-2} + \xi \sum_{j \geq 2} \frac{r_j (iz)^j}{j!} \varepsilon^{j-1} \right\} = \sum_{j \geq 0} T_j(z, \xi) \varepsilon^j. \quad (5.7)$$

Then $P_j(\xi)$ is the integral of T_j against $e^{-vz^2/2 + ir_1\xi z}$, divided by the leading Gaussian integral. Only finitely many terms are needed for each j . As a consistency check, the fixed- m coefficient in the manuscript is

$$\beta_m = \beta_0 + m \left[-\frac{r_1 + r_2/2}{v} + \frac{\kappa_3 r_1}{2v^2} \right] - \delta_\Gamma m^2.$$

The quadratic term is exactly the first Taylor coefficient of the Gaussian damping factor. The uniform theorem resums that part when m^2/k is no longer negligible.

5.1 Uniform optimal truncation with a growing reflected exponent

Theorem 5.2 (A moving-exponent half-term law). *Fix $L, E < \infty$. Let $n, N \rightarrow \infty$ through positive integers, $0 \leq m \leq L\sqrt{N}$, and $|\eta| \leq E$, where $\eta = N\Lambda - n$. Then*

$$\frac{M_{n,m}/n! - \sum_{k=1}^{N-1} A_{k,m}/k^n}{A_{N,m}/N^n} = \frac{1}{2} + \frac{3/2 - \delta_\Gamma m^2/N - \eta}{4N} + O_{L,E}(N^{-3/2}). \quad (5.8)$$

The denominator is nonzero for all sufficiently large N , uniformly in the stated range. In particular the error is asymptotic to half the first omitted term, and its eventual sign is $(-1)^{N+m-1}$. The theorem supplies a proved growing range; it does not assert this range is maximal.

Proof. We give the contour and tail details because neither a formal growing substitution in the fixed- m formula nor an infinite divergent Taylor tail would justify this statement. Let h be the inverse germ $h = q\phi(h)$, let $g(t) = -h(-t)$, and set

$$W_m(x) = \int_0^x B(u)^m du, \quad G_m(t) = W_m(g(t)).$$

Lagrange inversion gives $G_m(t) = \sum_{k \geq 1} A_{k,m} t^k$ near zero. The manuscript's inverse-gamma argument shows that g continues to a slit disk of radius greater than ρ , whose only singularity on its circle of convergence is at $-\rho$. For completeness, positivity of the coefficients of h gives $|h(q)| < \tau$ at every $|q| = \rho$, $q \neq \rho$; then $|q\phi'(h(q))| < \rho\phi'(\tau) = 1$ allows analytic continuation. At $q = \rho$ the first derivative of $u/\phi(u)$ vanishes and the second does not, so ordinary local inversion gives a convergent

square-root expansion. Compactness produces a common slit disk. Shrink it so $|g| < r_0 < 1$ there; this makes composition with every W_m valid.

Choose a fixed outer radius $R > \rho$ in that disk and a small $0 < \epsilon < \Lambda$ with $\rho e^\epsilon < R$. Along its negative cut define the real, possibly signed density

$$\nu_m(s) = \frac{(-1)^m}{2\pi i} (G_m(-s + i0) - G_m(-s - i0)), \quad \rho < s < R.$$

Put $\mathcal{B}_{N,m} = (-1)^{N+m-1} A_{N,m}$. Cauchy contour deformation yields

$$\mathcal{B}_{N,m} = I_{N,m} + O(M^m R^{-N}), \quad I_{N,m} = \int_\rho^R s^{-N-1} \nu_m(s) ds, \quad (5.9)$$

for a fixed M , with an inessential enlargement of the constant. The corresponding remainder contour has the additional kernel $s/(s+t)$. The common factor and sign cancel after division by $A_{N,m} t^N$. Outer-contour errors, including this division, are exponentially small for $m \leq L\sqrt{N}$ and $\rho e^{-\epsilon} \leq t \leq \rho e^\epsilon$.

The jump need not be positive in this moving range. The needed absolute bound is as follows. Write $u = (s - \rho)/\rho$. The square-root inverse has $g(-s \pm i0) = -\tau + O(\sqrt{u})$ and a connecting path of length $O(\sqrt{u})$. On that path $|B(z)| \leq b \exp(C_0 \sqrt{u})$ for sufficiently small fixed u . Since $W'_m = B^m$,

$$|\nu_m(\rho(1+u))| \leq C_1 b^m \sqrt{u} \exp(C_0 m \sqrt{u}).$$

On the rest of the finite cut it is at most $C_2 M^m$. Substituting $w = Nu$, using $m \leq L\sqrt{N}$ and $(1+u)^{-N-1} \leq e^{-cNu}$ on the small interval, proves for every fixed $j \geq 0$

$$\int_\rho^R \left| \frac{s-\rho}{\rho} \right|^j s^{-N-1} |\nu_m(s)| ds \leq C_{L,j} b^m \rho^{-N} N^{-j-3/2}. \quad (5.10)$$

The integrable majorant is a constant times $w^{j+1/2} e^{-cw+C_0 L \sqrt{w}}$. By Theorem 5.1, division by $\mathcal{B}_{N,m}$ turns (5.10) into $O_{L,j}(N^{-j})$. This absolute estimate prevents cancellation in the signed jump from invalidating the Taylor-remainder bounds.

The first normalized cut moment is computed without expanding the moving jump explicitly. The exact finite-cut identity gives

$$\int_\rho^R \frac{s-\rho}{\rho} s^{-N-1} \nu_m(s) ds = \rho^{-1} I_{N-1,m} - I_{N,m}.$$

Use Theorem 5.1 through order N^{-1} , with relative remainder $O_L(N^{-3/2})$, both at $N-1$ and at N with the same m . The polynomials in that expansion are smooth for bounded $m/\sqrt{N}$. Their ratio changes by $O_L(N^{-3/2})$, while the leading factors give

$$\begin{aligned} \frac{\mathcal{B}_{N-1,m}}{\rho \mathcal{B}_{N,m}} &= \left(\frac{N}{N-1} \right)^{3/2} \exp \left(-\frac{\delta_\Gamma m^2}{N(N-1)} \right) (1 + O_L(N^{-3/2})) \\ &= 1 + \frac{3/2 - \delta_\Gamma m^2/N}{N} + O_L(N^{-3/2}). \end{aligned}$$

For $r = t/\rho$,

$$\frac{s}{s+t} = \frac{1}{1+r} + \frac{r}{(1+r)^2} u + O(u^2)$$

uniformly on the fixed middle interval. Equations (5.9)–(5.10) therefore give

$$\frac{G_m(t) - \sum_{k < N} A_{k,m} t^k}{A_{N,m} t^N} = \frac{1}{1+r} + \frac{r}{(1+r)^2} \frac{3/2 - \delta_\Gamma m^2/N}{N} + O_L(N^{-3/2}). \quad (5.11)$$

It remains to justify averaging this local statement. The exact layer-cake identity is

$$\frac{M_{n,m}}{n!} = \frac{1}{\Gamma(n)} \int_0^\infty y^{n-1} G_m(e^{-y}) dy.$$

After subtracting the finite sum, the ratio in (5.8) is the expectation of the left side of (5.11) at $t = e^{-Y}$, where Y has gamma shape n and rate N . Its probability outside $|Y - \Lambda| \leq \epsilon$ is $O(e^{-cN})$ for bounded η , by the elementary gamma moment-generating function. Uniform subexponential bounds on the normalized remainder suffice on that complement. To obtain them, Cauchy's coefficient formula on $|u| = \tau$ gives

$$|A_{k,m}| \leq \frac{\tau}{k} \rho^{-k} H^m$$

with a fixed H . Below $t = \rho e^{-\epsilon}$ sum the absolutely convergent Taylor tail. Above $t = \rho e^\epsilon$ bound the finite polynomial by its absolute terms. Division by the uniform lower bound for $|A_{N,m}|$ in Theorem 5.1 gives at most a polynomial in N times $e^{C_L \sqrt{N}}$ in both cases. The remaining function term on the upper interval satisfies $0 \leq G_m(t) \leq m!$ for $0 \leq t \leq 1$: indeed $0 \leq g(t) \leq 1$ and $B(x) \leq -\log(1-x)$, since $\log \Gamma(2-x) \leq 0$ by log convexity between one and two. Thus $W_m(g(t)) \leq \int_0^1 [-\log(1-x)]^m dx = m!$. After division its size is at most a polynomial times $e^{-\epsilon N} m! b^{-m}$, hence exponentially small in N because $m \log(m+1) = O_L(\sqrt{N} \log N)$. The weighted outer contribution is therefore exponentially small.

Finally put $V = Y - \Lambda$. Then

$$\mathbb{E}V = -\eta/N, \quad \mathbb{E}V^2 = O_E(N^{-1}), \quad \mathbb{E}V^3 = O_E(N^{-2}), \quad \mathbb{E}V^4 = O_E(N^{-2}).$$

The leading kernel has expansion $1/(1+e^{-V}) = 1/2 + V/4 - V^3/48 + O(V^4)$, so its expectation is $1/2 - \eta/(4N) + O_E(N^{-2})$. The next kernel is $e^{-V}/(1+e^{-V})^2 = 1/4 + O(V^2)$, since its derivative vanishes at zero. Its expectation is $1/4 + O_E(N^{-1})$. Averaging (5.11) gives (5.8). The signs and nonvanishing follow from Theorem 5.1. $\square$

What the theorem resolves and what it leaves open. The fixed-reflected-exponent warning in the manuscript is correct; it should now be supplemented by a separate theorem for $m = O(\sqrt{N})$. The square-root scale is the first scale on which the leading late residue changes by a nontrivial factor. It does not destroy the half-term law, but it changes that law's first correction. The fixed- m correction $\beta_m/(4N^2)$ contains the same term $-\delta_\Gamma m^2/(4N^2)$; the result above proves the relevant resummation uniformly instead of substituting a growing parameter in an uncontrolled remainder.

The next subsection proves a small proportional range for the coefficients themselves. The maximal extent of that range and extensions of the half-term remainder law are separate questions: a coefficient asymptotic alone does not control a normalized signed-cut remainder.

5.2 A small proportional range and a second coefficient transition

The preceding circle argument also gives a nonzero proportional range for the late coefficients themselves. This strengthens the Gaussian formula without changing the stated range of the half-term remainder law.

Theorem 5.3 (Uniform small proportional saddle). *There exists $c_0 \in (0, 1)$ with the following properties. For $0 \leq c \leq c_0$ there is a unique real-analytic continuation τ_c of $\tau_0 = \tau$ solving*

$$D \log \phi(\tau_c) + cD \log a(\tau_c) = 1. \quad (5.12)$$

Its curvature

$$V(c) = D^2 \log \phi(\tau_c) + cD^2 \log a(\tau_c)$$

is positive. Define

$$S(c) = \log \phi(\tau_c) + c \log a(\tau_c) - \log \tau_c, \quad C(c) = \frac{\tau_c}{\sqrt{2\pi V(c)}}. \quad (5.13)$$

Uniformly for integers $0 \leq m \leq c_0 k$, with $c = m/k$,

$$A_{k,m} = (-1)^{k+m-1} C(c) k^{-3/2} e^{kS(c)} (1 + O(k^{-1})). \quad (5.14)$$

There is a uniform expansion to every fixed order in integer powers of k^{-1} , with real-analytic coefficient functions of c . The constant c_0 is asserted to exist; no numerical lower bound for it or maximality assertion is included.

Proof. Use the logarithmic radius $x = \log u$ and put $f(x) = \log \phi(e^x) - x$, $g(x) = \log a(e^x)$ locally near $x_0 = \log \tau$. Then $f'(x_0) = 0$ and $f''(x_0) = v > 0$. The analytic implicit-function theorem applied to $f'(x) + cg'(x) = 0$ supplies $x(c) = \log \tau_c$ near zero. After shrinking the interval, τ_c lies in a fixed compact $I \subset (0, 1)$ where a is positive, and $V(c) > v/2$. Uniqueness here means uniqueness of this local continuation; it does not classify other real or complex saddles.

Choose a small fixed angular neighborhood $|t| < \epsilon$ on which $a(ue^{it})$ is nonzero for every $u \in I$. Positivity of the Taylor coefficients of ϕ gives a strict uniform bound

$$q := \sup_{u \in I, \epsilon \leq |t| \leq \pi} \frac{|\phi(ue^{it})|}{\phi(u)} < 1.$$

Also let

$$H = \max \left\{ 1, \sup_{u \in I, |t| \leq \pi} \frac{|a(ue^{it})|}{a(u)} \right\} < \infty.$$

Further decrease c_0 so that $qH^{c_0} < 1$. In the Cauchy integral on $|u| = \tau_c$, the outer arc divided by the positive saddle amplitude is at most $q^k H^m \leq (qH^{c_0})^k$. This is the global estimate needed for proportional m : no logarithm or fractional power of a on the full circle is assumed.

On the small arc the normalized local logarithmic phase is

$$f(x(c) + it) - f(x(c)) + c(g(x(c) + it) - g(x(c))).$$

Its linear term vanishes by (5.12); its quadratic term is $-V(c)t^2/2$. Analyticity, reality on the real radius, and compactness give a uniform Gaussian modulus bound for a sufficiently small fixed ϵ . Choose this ϵ first and then make the preceding final reduction of c_0 , so the outer-arc estimate uses the same angular neighborhood. Taylor expansion with $t = z/\sqrt{k}$ and amplitude e^{it} now gives

$$\int e^{k\{f(x(c)+it)-f(x(c))+c[g(x(c)+it)-g(x(c))]\}} e^{it} dt = \sqrt{\frac{2\pi}{kV(c)}} (1 + O(k^{-1})).$$

The discarded arcs are exponentially small. Odd terms integrate to zero, so only integer powers of k^{-1} occur. Analyticity and uniform derivative bounds also prove the asserted all-orders expansion. Restoring the Cauchy prefactor $(-1)^{k+m-1} \tau_c e^{kS(c)} / (2\pi k)$ proves the theorem. $\square$

Corollary 5.4 (Explicit rate expansion and cubic transition). *Put*

$$d_1 = -\frac{r_1 + r_2/2}{v} + \frac{\kappa_3 r_1}{2v^2},$$

$$d_3 = \frac{r_2 r_1^2}{2v^2} - \frac{\kappa_3 r_1^3}{6v^3}.$$

Then

$$S(c) = \Lambda + c \log b - \delta_\Gamma c^2 + d_3 c^3 + O(c^4),$$

$$C(c) = C(1 + d_1 c + O(c^2)). \quad (5.15)$$

Consequently the Gaussian-resummed leading expression

$$A_{k,m} \sim (-1)^{k+m-1} C b^m \rho^{-k} k^{-3/2} e^{-\delta_\Gamma m^2/k} \quad (5.16)$$

is uniform for $m = o(k^{2/3})$: precisely, it holds uniformly for $0 \leq m \leq \varepsilon_k k^{2/3}$ whenever $\varepsilon_k \rightarrow 0$. For each fixed $L < \infty$, uniformly for $0 \leq m \leq Lk^{2/3}$,

$$A_{k,m} = (-1)^{k+m-1} C b^m \rho^{-k} k^{-3/2} \exp\left(-\frac{\delta_\Gamma m^2}{k} + \frac{d_3 m^3}{k^2}\right) (1 + O_L(k^{-1/3})). \quad (5.17)$$

In particular, if $m/k^{2/3} \rightarrow \chi \geq 0$, the ratio to the right-hand leading expression in (5.16) tends to $e^{d_3 \chi^3}$. The numerical value is $d_3 = -0.00545592970200504\dots$; the exact defining formula is used in the theorem.

Proof. Differentiating the saddle equation at zero gives $x'(0) = -r_1/v$. Since $S(c) = f(x(c)) + cg(x(c))$ and $f'(x(c)) + cg'(x(c)) = 0$, expansion through degree three gives

$$S(c) = \Lambda + c \log b - \frac{r_1^2}{2v} c^2 + \left(\frac{r_2 r_1^2}{2v^2} - \frac{\kappa_3 r_1^3}{6v^3}\right) c^3 + O(c^4).$$

For the prefactor,

$$(\log C)'(0) = x'(0) - \frac{\kappa_3 x'(0) + r_2}{2v} = d_1.$$

This proves (5.15). Upon setting $c = m/k$, Theorem 5.3 becomes

$$A_{k,m} = (-1)^{k+m-1} C b^m \rho^{-k} k^{-3/2} e^{-\delta_\Gamma m^2/k} \exp\left(\frac{d_3 m^3}{k^2} + O\left(\frac{m^4}{k^3}\right)\right)$$

$$\times \left(1 + \frac{d_1 m}{k} + O\left(\frac{m^2}{k^2} + \frac{1}{k}\right)\right).$$

For $m = o(k^{2/3})$ all corrections on the last two factors tend to one uniformly. For $m \leq Lk^{2/3}$, the quartic rate error and the first prefactor correction are $O_L(k^{-1/3})$, which gives (5.17) and its limiting assertion. Finally, when $m = \xi\sqrt{k}$, the prefactor and cubic rate contribute $(d_1 \xi + d_3 \xi^3)/\sqrt{k}$. This is exactly $P_1(\xi)$ in (5.5), independently matching the square-root calculation. $\square$

Proposition 5.5 (Strict cubic damping). *The exact cubic rate coefficient satisfies $d_3 < 0$. Thus the transition factor $e^{d_3 \chi^3}$ is strictly below one for every $\chi > 0$.*

Proof. We already know $\tau > 1/2$ and $r_1 < 0$. The positive log-gamma series also gives

$$D \log \phi(3/5) > \frac{1}{2} \frac{3}{5} + \frac{(3/5)^2}{1 - 3/5} = \frac{6}{5} > 1,$$

so $\tau < 3/5$. Put $P = \psi(1 + \tau)$ and $Q = \psi'(1 + \tau)$. The trigamma series and its derivatives [7] show that ψ is increasing and concave and that

$$Q = \sum_{n \geq 0} \frac{1}{(1 + \tau + n)^2} > \frac{1}{1 + \tau}.$$

Using $\gamma > 1/2$, $\log 2 > 2/3$, and $\pi^2 < 10$ in the classical values at $3/2$ gives

$$0 < P < \psi(3/2) + \frac{1}{10} \psi'(3/2) < \frac{1}{6} + \frac{1}{10} = \frac{4}{15}, \quad \tau Q > \frac{\tau}{1 + \tau} > \frac{1}{3}.$$

In particular $P < \tau Q$.

Write $T = \psi'(1 - \tau) > 0$. Termwise comparison in the polygamma series yields $-\psi''(1 - \tau) \leq 2T/(1 - \tau)$. At the saddle,

$$v = 1 + \tau^2 T, \quad \kappa_3 = 1 + 3\tau^2 T - \tau^3 \psi''(1 - \tau).$$

Since $\tau < 3/5$, these imply $\kappa_3 < 1 + 6\tau^2 T < 6v$, or $\kappa_3/(3v) < 2$. Finally,

$$r_1 = -\frac{\tau P}{b}, \quad r_2 = r_1 - r_1^2 - \frac{\tau^2 Q}{b}.$$

Because $r_1 < 0$,

$$r_2 - \frac{\kappa_3 r_1}{3v} < -r_1 - r_1^2 - \frac{\tau^2 Q}{b} = \frac{\tau}{b} (P - \tau Q) - r_1^2 < 0.$$

Multiplication by $r_1^2/(2v^2) > 0$ proves the assertion. $\square$

The proportional theorem settles existence of a small positive sector. The remaining question is its maximal extent: determine when another arc or saddle becomes competitive and how that affects coefficient signs. The cut-density estimates in the half-term theorem normalize by an exponentially damped coefficient and were proved only for bounded $m/\sqrt{N}$; the new coefficient theorem alone does not enlarge that remainder theorem.

6 Audit, corrections, and verification

The source audit was performed against the exact commit specified on the title page. It was selective and mathematical: we checked the definitions and proof dependencies needed for the four continuations, then compared the conclusions with the manuscript's explicit open questions. This is not an assertion that every formula in the entire repository has been independently re-proved.

6.1 A wording correction concerning non-elementarity

The paragraph preceding Equation `vertical:eq:Im3` in `chapters/03-algebraic.tex` describes the imaginary part of $\text{Li}_3(1 + i)$ as *no longer elementary at all*. The displayed hypergeometric representation does not establish that assertion. An exact expression in terms of a central-binomial tower constant is an evaluation in a specified language. It is not a proof that the value cannot

belong to some smaller algebra of constants. Indeed, without specifying the meaning of elementary for a number, the assertion has no definite mathematical test.

A precise replacement is:

The imaginary part at $1 + i$ is represented here by the central-binomial Gaussian tower constant:

followed by the existing, unchanged equation, and then:

This representation does not establish non-reducibility to a specified smaller algebra of constants.

The preceding claim that the general real part *does not close* should similarly be scoped to the computation being described. One can say that differentiating and integrating the displayed formula introduces an inverse-tangent-square integral, and that no general reduction of that integral is supplied there. This is a correction to the strength of the prose; the displayed special-value formulas are not contradicted by this audit. An exact-context patch implementing these two edits is included in `integration/non_elementarity_wording.patch`.

6.2 Statements that should be updated after integration

Uniform Euler error. The warning that a bound on $|g_{a,b}|$ is insufficient to control $2^N(E_N - g_{a,b})$ was correct. The explicit calculation in Section 2 shows $R_2(1, 1) > R_1(1, 1)$. Theorem 2.4 now supplies the missing uniform comparison. The open-status sentences in the subcritical and real-order Euler sections should therefore be replaced by a reference to the new theorem. The sharper constant 1 on the restricted domain $a \geq 1$ remains useful and should be retained. A global budget of 57/50 is convenient, but it is not the sharp bound on every subdomain.

Quartic threshold. The final paragraph of `chapters/05-real-turning.tex` correctly labels its approximate simultaneous zero as diagnostic. Our exact rectangle and derivative certificate promote that particular location, its local uniqueness, and its sixth-order behavior to theorems. They do not establish uniqueness on all $0 < b < 1$. The current existence proof based on opposite signs of the quartic coefficient remains a useful conceptual introduction. It should be followed by the new local classification and explicit rational two-extremum example, with a separate global open question.

Reflected coefficients and residues. The coefficient $A_{k,m}$ defined by Lagrange inversion and the meromorphic residue are related by

$$\operatorname{Res}_{z=k} \int_0^1 \Gamma(x)^z \log^m \Gamma(1-x) dx = -k A_{k,m}.$$

Any compact description of signs should retain this factor. The moving-parameter theorems extend the fixed- m estimates; they do not justify inserting an arbitrary growing m into a fixed-parameter error term. The moment transition governed by $n/(m+1) - \log m$ is a different simultaneous limit and should remain separately indexed.

Height-one reductions and the literature. The Nielsen-polylogarithm work of Charlton, Gangl, and Radchenko is a necessary part of the attribution for Section 3. In particular, the weight-seven height-one reduction is an established functional equation. Our presentation supplies a directly replayable shuffle certificate, an explicit product correction, and a comparison with the full

binary quotient. The finite-direction formula is stated in that full quotient, with its precise relation space visible. Even when a formal obstruction is nonzero, it must not be described as a proof that a complex number cannot reduce.

Unresolved evaluations. The S_6 and revised S_8 candidates in the existing discovery ledger remain conjectural. Neither the new Euler budget, a very small residual, nor a formal obstruction to one reduction method proves or disproves these special-value identities. This article does not alter their evaluation status.

6.3 The finite arithmetic proofs

The exact programs use Python integers and `fractions.Fraction`. No binary floating-point comparison decides the sign of an interval in these proofs. The use of Python is an implementation choice, not a trust claim about all software: the written formulas specify finite statements that can be replayed in another exact-arithmetic environment. These are computer-assisted arithmetic proofs, not proof-assistant formalizations.

Verifier	Exact task	Logical role
<code>verify_euler.py</code>	Thirty-one rational axis enclosures, the rational global budget, selected rational-order enclosures, and corrupted candidate controls.	Proves the finite inequalities used in Proposition 2.6.
<code>verify_depth_orbits.py</code>	Exact Lyndon-bracket pairings, word-rank calculations, the weight-seven product expansion, and the weight-eight annihilator.	Finite proof certificates for the explicit identities; finite corroboration of the all-weight rank theorem.
<code>certify_radial_degeneracy.py</code>	Outward intervals for the simultaneous-zero rectangle, the endpoint signs, the Jacobian, and the sixth coefficient.	Supplies the finite sign claims needed for the analytic existence and unfolding proof.
<code>certify_two_extrema.py</code>	Traps the actual implicit zero; bounds the infinite series and its mixed derivatives; proves three signs and uniform strict concavity.	Proves the two-extremum statement for the exact rational parameter pair.

The Euler verifier also checks 6120 rational kernel instances. That grid is a useful regression check; the all- N kernel theorem is proved by the probability-mixture argument, not by extrapolation from the grid. Its two deliberately false controls are a proposed global constant 1 and the changed finite value $E_2(1,1) = -1/7$. The first is rejected by a certified analytic enclosure; the second is rejected by exact rational evaluation, which gives $-1/8$.

The word-rank checks through weight ten cover 25 weight-depth cases for each of the six-map and full-linear spans, and 34 additional finite-direction cases. They are deliberately computed from the expanded free-Lie words rather than by reusing the dimension formula as the answer. The height-one membership checks also test the asserted threshold on individual words. The weight-eight obstruction is checked on 222 transformed seed words and 1792 shuffle products. The weight-seven product correction contains 864 collected monomials; expanding their shuffle products reproduces the residual word combination exactly.

N	n	m	Observed normalized remainder	First-correction prediction
24	30	5	0.5127180890556	0.5116978569285
24	30	15	2.1726463253744	0.5111559565783
48	61	7	0.5094463157920	0.5091282364092
48	61	21	0.6787471547306	0.5088627052376
96	121	10	0.5027543017622	0.5026340767530
96	121	29	0.5028409944568	0.5025085929532

Table 1: Diagnostic values for the left side of (5.8), compared with $1/2 + (3/2 - \delta_{\Gamma} m^2 / (N - \eta)) / (4N)$. All six rows are retained, including the substantial preasymptotic deviations at larger m and smaller N . The observed values are high-precision quadrature results, not interval enclosures.

The first radial certificate uses a 2^{-192} outward grid. Its logarithms are bounded by an atanh series with an explicit geometric remainder, and exponentials by a positive Taylor sum and a geometric tail. The second certificate range-reduces logarithms before applying the same principle. The programs record both the raw rational intervals and the simpler rational enclosures printed in the theorems. Very small positive or negative quantities are therefore proved to have their asserted signs; their size is not inferred from decimal precision alone.

6.4 Numerical diagnostics and independent review

The optional numerical programs require `mpmath`, and the asymptotic symbolic checks additionally use `sympy`. They serve a different purpose from the exact programs. The complete weight-seven functional identity, including endpoint constants and products, was checked by direct disk series at $z = 1/4$ and $z = 1/3$ with 80 working decimal digits. The discrepancies were approximately $3.7 \cdot 10^{-80}$ and $2.1 \cdot 10^{-80}$. The proof comes from the exact word identity and the branch argument; these numerical values check their implementation.

The proposed sharp rate for $C_N - 1$ is accompanied by 210-digit axis calculations and a few positive- a checks. The latter replace $g_{a,b}$ by E_{512} , with analytic contribution to the scaled truncation error at most $(5/4)2^{N-512}$. That proved truncation bound is separate from floating-point error. Neither these values nor the numerical axis stationary points certify a global maximum.

For the reflected coefficients, the diagnostic data use 180 working digits, compare two cases with independent Cauchy quadrature, and repeat the six direct scaled-moment integrals at 215 digits. The repeated moment ratios differed by less than $1.5 \cdot 10^{-108}$ in those cases. These are high-precision agreements, not rigorous interval error bounds. The accompanying tables retain examples outside the visible asymptotic regime, including large normalized deviations at moderate indices. Suppressing those rows would obscure the meaning of a uniform theorem whose constants depend on a fixed range parameter.

The four main arguments received separate mathematical cross-checks during preparation. In particular, the endpoint corrections in the Nielsen identity, the coefficient-dual normalization in the binary-form calculation, the signed-cut normalization in the half-term theorem, and the complete implicit-derivative certificate for the explicit radial example were checked independently of their initial derivations. This record describes the scope of review; it is not a substitute for external peer review.

6.5 Reproduction and integration

The archive includes a single entry point, `code/reproduce.py`, which runs the exact verifiers without requiring access to the original repository. Optional switches replay the numerical diagnostics and regenerate the figures. The figures are explanatory numerical plots; their captions identify the exact statements that supply proof. Run `latexmk-pdfarticle.tex` from the package root to rebuild the document using the supplied section and figure files.

The provenance directory records the pinned commit, the relative source paths used in the audit, and their SHA-256 digests. The integration guide maps each section and certificate to the appropriate manuscript chapter. It also records the small preamble requirements and label prefixes, so that the standalone article can be integrated without depending on an unstated local macro package. No changes were applied to the source repository.

7 Further research: precise questions and next steps

The results above close several concrete gaps, but each also identifies a natural boundary. The following programme separates conjectures with specific existing evidence from broader questions. The proposed tasks are intended to lead to additional theorems or falsifiable identities, not merely more decimal tables.

7.1 Euler bounds: rates, extremizers, and further moment families

Sharp bounds for a fixed truncation index. The global constant C_* is forced by $N = 1$ and $a \downarrow 0$. For practical computations with large N , it leaves substantial room. Define the exact pointwise envelope

$$K_N(v) = \max_{0 \leq x \leq 1} \{W_N(vx) - W_N(x)\}, \quad 0 < v < 1.$$

Theorem 2.3 determines the supremum over all N , but does not determine $K_N(v)$ in a useful closed form for $N \geq 2$. Find its maximizing point, its dependence on N , and the strongest integrated bound it yields for fixed a, b . A derivative equation for the maximizing point is elementary, so the challenge is an interpretable exact inequality rather than a numerical maximization.

The scaling $x = y/N$ suggests the limiting comparison $e^{-vy} - e^{-y}$, whose maximum is $(1 - v)v^{v/(1-v)}$. This suggests asking for an error-controlled limit of $K_N(v)$, including its behavior as $v \downarrow 0$ and $v \uparrow 1$. Uniformity at those endpoints matters before integrating against an order-dependent measure. The elementary limiting calculation alone does not supply that uniformity.

Fractional-order late Euler errors. The integer-order manuscript already gives a polynomial in $\frac{1}{2} \log N$ multiplying $2^{-N} N^{-1/2}$, with leading coefficient determined by the total weight. Extend the full expansion to positive nonintegral a, b , explicitly tracking the extra endpoint terms that appear when an exponent crosses an integer. In particular, establish a uniform version of the fixed-order leading behavior suggested by the two-measure convolution,

$$E_N - g_{a,b} \sim \frac{\sqrt{\pi}}{2^{a+b}\Gamma(a+b)} \frac{2^{-N}(\log N)^{a+b-1}}{\sqrt{N}}.$$

For positive integral orders this is already a theorem in `chapters/05-signed-kernels.tex`; the research target is the fractional continuation and its uniform error structure. A proof should separate the region where both logarithmic moment variables are large from the boundary layers in which only one is large. The signed-measure representation can fail to have finite mass, so the positive two-measure difference is a promising starting point.

The rate of approach to the asymptotic global constant. Theorem 2.11 proves $C_N \rightarrow 1$, and Proposition 2.12 gives a concrete lower asymptotic. The natural next target is the matching upper bound.

Conjecture 7.1 (Asymptotic optimal Euler constant). *With*

$$p = \frac{\log 2}{\log(3/2)}, \quad q = \frac{\log 3}{\log 2}, \quad c = (1 - q^{-1})q^{-p},$$

the optimal scaled-error constant satisfies

$$C_N - 1 \sim cN^{-p}.$$

The matching lower bound is proved. The global upper bound over positive a is open, and even the sharp upper asymptotic for the supremum over all b on the axis requires an argument. The moving-order phase law supplies a broad region with $R_N \rightarrow 1$, but that limit does not distinguish errors of order N^{-p} . Any statement about near maximizers must specify the accuracy: an error $o(C_N - 1)$ is relevant here, whereas an error merely $o(1)$ need not force $a \rightarrow 0$. Monotonicity of C_N is also not asserted.

The diagnostic program `code/axis_remainder_diagnostics.py` evaluates finite Euler sums at 210 decimal digits. For the following indices it finds numerical axis stationary points, with the displayed scaled excess:

N	$\hat{b}_N$	Stationary b	$(R_N(0, b) - 1)/(cN^{-p})$
4	4.5549051506	4.7024351462	0.8809410
16	7.9739277333	7.8500957399	1.0384240
64	11.3929503160	11.2260085539	1.0760668
256	14.8119728987	14.6711951089	1.0684372

Their global optimality is not certified. At $b = \hat{b}_N$, the three tested positive values $a = 10^{-4}, 10^{-2}, 10^{-1}$ all give smaller remainders than $a = 0$. These are limited axis and local checks; they do not establish the conjecture or monotone convergence of the displayed ratios. An effective compact-rectangle estimate, combined with a sharp analysis of the axis scaling, is one possible route to an upper bound.

Further moving-order boundaries. Theorem 2.10 treats two orders growing proportionally to $\log N$. Determine the finer behavior when one of the leading coefficients A, B tends to zero, or when one order remains fixed while the other crosses the logarithmic threshold. The bounded-width gamma variable then interacts with the smoothed threshold itself, and the two normal profiles need not describe the first correction. Also study orders tending to zero, where endpoint concentration affects the fractional late-error expansion. A useful result would connect the several scaling regimes with explicit overlapping error bounds.

Other moment products. Theorem 2.8 applies to any two finite positive measures. Explore shifted harmonic numbers, Hurwitz-type inner sums, mixtures of orders, and products arising from completely monotone sequences. For each family, identify the sharp measure-specific constant rather than automatically using $(1 + \sqrt{2})/2$. Higher-depth nested partial sums lead to higher divided differences of the same kernel. Determine which of those differences have a fixed sign and what sharp moduli replace the two-point inequality. This is a concrete route toward certified Euler algorithms for real-order higher-depth polylogarithms.

7.2 Formal depth and genuine functional reduction

Exploit the full binary formula constructively. The finite-direction dimension theorem gives the rank but does not choose the best basis for an identity search. Construct rational bases adapted to the irreducible $\text{Sym}^{w-2j} V \otimes \det^j$ summands and their confluent interpolation conditions. This should make it possible to solve reduction problems without forming all 2^w words. The weight-seven product correction is explicit but lengthy; a basis respecting reversal, complementation, and endpoint regularization may produce a much shorter analytic expression.

Compare with the Nielsen literature at each weight. The height-one results must be compared with the known Nielsen reductions, including their meaning modulo products and their exact symbol spaces. A useful next deliverable is a table that gives, for every weight in a practical range, the published reduction, a word certificate in the repository’s conventions, and the endpoint correction on a stated branch domain. It should identify equivalence of relations, not count the same identity multiple times because of a change of notation.

Go beyond three punctures. The six transformations act on only three projective letter directions. A general rational substitution may introduce additional singularities and therefore a larger alphabet; it is not represented by an arbitrary 2×2 change of the old letters. Extend the rank problem to a specified puncture set and a specified collection of rational maps. Describe the kernel of specialization back to the original periods. The weight-eight obstruction here is a useful test case: which genuinely new pullbacks or distribution relations remove it, and what new constants do they introduce?

Separate formal and period questions. For the unresolved Gaussian S_6 and S_8 evaluations, specify the expected algebra of constants before searching. Construct the corresponding functional relation space, map it to the target specialization, and compare the proposed identity with that image. A formal failure would indicate that the chosen tools are insufficient. It would not prove period independence. Conversely, an exact functional certificate with controlled endpoint constants would settle an evaluation without relying on an integer-relation residual.

7.3 The global radial phase portrait

Let $a_*(b)$ be the already proved unique solution of $K = 0$ in $(1, 2)$ for $0 < b < 1$, and put $q(b) = Q(a_*(b), b)$. The pinned manuscript proves opposite signs at two rational parameters and reports a wider diagnostic scan. The exact certificate here proves one simple zero near the upper endpoint of that interval.

Conjecture 7.2 (Unique quartic transition on the quadratic threshold). *The function q has exactly one zero on $0 < b < 1$, namely the certified b_+ . It is negative for $0 < b < b_+$ and positive for $b_+ < b < 1$.*

The local sign at the certified zero is proved, and the existing finite scan supports the global statement. Neither fact excludes additional zeros away from the isolating box. A rigorous approach should combine asymptotic analysis as $b \downarrow 0$ and $b \uparrow 1$ with a finite interval cover of the remaining compact region. Direct interval evaluation may be inefficient near $b = 1$, where several low-order coefficients nearly cancel. Factoring the known degeneration at $(a, b) = (1, 1)$ before bounding remainders is likely to be essential.

Global continuation of the two-extremum region. The local fold and the explicit rational example prove that a minimum followed by a maximum can occur. Follow the two critical branches to larger radii and determine how they end: collision, escape through $\rho = 1$, or interaction with another branch. Can more than two radial critical points occur for some positive orders? The local concavity proof does not address this global question. A plausible first step is an interval continuation of the coupled equations for the angular zero and its radial derivative, with a separate proof that no branch is missed.

The older integer-order monotonicity question. The conjectured full-radius decrease for integer $a \geq 2$, and the special $a = 1$ behavior, remain interesting. The fractional examples do not refute them: their parameters are close to, but different from, $(1, 1)$. Look for a sign-definite integral expression for the radial derivative in the integer range. A proof based only on the coefficient K would establish local behavior and leave the rest of the radius interval untreated.

Higher degeneracies. The binomial implicit equation gives every radial coefficient without guessing higher Chebyshev expansions. Search for families with $K = Q = R = 0$ after adding a deformation parameter, such as a Lerch shift or an extra inner moment. If a first nonzero coefficient appears at $\rho^{2\ell}$, generic one-parameter perturbation suggests turning radii with power $1/(2\ell - 2)$. State the transversality hypotheses explicitly and prove the local normal form before drawing a global bifurcation diagram.

7.4 Growing reflected exponents and optimal truncation

The maximal proportional-exponent regime. The shifted-saddle continuation establishes a nonzero interval of proportional growth. Determine the largest admissible interval by analyzing the real saddle equation and competing complex saddles. Positivity of the Taylor coefficients of $\Gamma(1 - u)$ is available, but $-\log \Gamma(1 + u)$ does not have a positive coefficient sequence. A local stationary point alone therefore does not prove global contour dominance for all m/k . Locate the first loss of dominance or the first saddle collision, and give an exact or certified description of the transition.

Move the optimal truncation index as well. When m/N is no longer small, the exponential rate of the coefficients changes. The optimal truncation index must then be determined from that joint rate, rather than retaining $N\Lambda \approx n$ without adjustment. Derive the correct stationarity equation when n, m, N grow together, and test whether a half-term law persists. The signed jump on the negative cut requires absolute moment estimates; eventual coefficient signs do not by themselves make the cut density positive.

Effective constants and preasymptotic behavior. Turn the present uniform big- O statements into explicit inequalities in a chosen practical range. The diagnostic examples show why this matters: a fixed range parameter can be mathematically admissible while its asymptotic behavior appears only at much larger indices. Effective contour radii, bounds on logarithmic derivatives, and a computable separation of the central arc from the outer contour would turn the theorem into a usable certified truncation prescription.

Connect the different moment scales. The coefficient scale $m \asymp \sqrt{k}$ and the direct-moment scale involving $n/(m + 1) - \log m$ arise from different representations. Determine their overlap after the optimal index is chosen, and compare the resulting asymptotic expansions term by term. A

matching theorem would explain which representation is stable in each region and where cancellations make one numerically preferable.

7.5 A practical order of attack

The most bounded next tasks are an exact global sign study for $q(b)$, a basis-efficient implementation of the finite-direction rank theorem, and effective constants for the shifted saddle. Each has a clearly specified output and can be checked against the certificates in this article. The longer projects are a global radial classification, an extension to larger puncture alphabets, and a joint optimal-truncation theory for proportional reflected exponents. The special-value conjectures should remain visible throughout, but their status should change only when an exact functional or arithmetic certificate becomes available.

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